

BANK OF THAILAND

Monetary Policy Report

January 2013

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
January 2013

Mr. Prasarn Trairatvorakul	Chairman
Mrs. Pongpen Ruengvirayudh	Vice Chairman
Mrs. Tongurai Limpiti	Member
Mr. Ampon Kittiampon	Member
Mr. Narongchai Akrasanee	Member
Mr. Siri Ganjarerndee	Member
Mr. Aswin Kongsiri	Member

Monetary Policy in Thailand

The Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprise of the Governor and two deputy Governors, as well as four distinguished external members representing various sectors of the country, with the aim to provide monetary policy decisions that are effective and transparent.

The Monetary Policy Objective

The MPC sets monetary policy with the objective to support sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

The Monetary Policy Target

The Cabinet approves the use of quarterly average core inflation with the range of 0.5-3.0 percent as its policy target for 2013, as proposed by the MPC together with the Minister of Finance. In the event that the target is missed, the MPC is required to explain the reasons and resolution thereof to the public.

The Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the key policy rate to signal the monetary policy stance.

Evaluating economic conditions and forecasts

The Bank of Thailand considers information from all sources, the macroeconomic model, data in each economic sector, as well as direct opinions from large enterprises and small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and covers all aspects, both in the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of Monetary Policy communication to the public, the MPC arrange many forms of communication, both in Thai and English, such as (1) organizing a press statement at 14.00 hrs. on the day of the meeting, (2) publishing Minutes of the MPC meeting two weeks after the meeting, and (3) publishing the Monetary Policy Report every quarter.

Monetary Policy Report

January 2013

Contents

1. Growth and Inflation Prospects and Monetary Policy 1

1.1 Growth and inflation prospects 1

1.2 Economic outlook 3

1.3 Inflation pressure 13

1.4 Monetary policy decisions 15

1.5 Appendix 17

BOX: Challenges from monetary easing by major economies and implications on Thailand 20

2. Recent Economic Developments 23

2.1 The global economy 23

2.2 The domestic economy 29

2.3 Costs and prices 37

BOX: Outlook on Japanese economic policy after the December 2012 election 40

3. Monetary and Financial Stability 43

3.1 Financial markets 43

3.2 Financial institutions 46

3.3 Non-financial sectors 48

BOX: The 300 baht minimum wage policy and its economic impact 55

Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy experienced better-than-expected growth in the second half of 2012 due to greater domestic demand momentum, while impacts from the slowdown in exports following global economic conditions were limited. Private consumption and investment were expected to exhibit steady growth over the projection period supported by a number of favorable factors. Meanwhile, exports recovered gradually and would lend more support to growth in the latter half of 2013. Inflation pressure was likely to remain subdued, similar to the previous assessment.

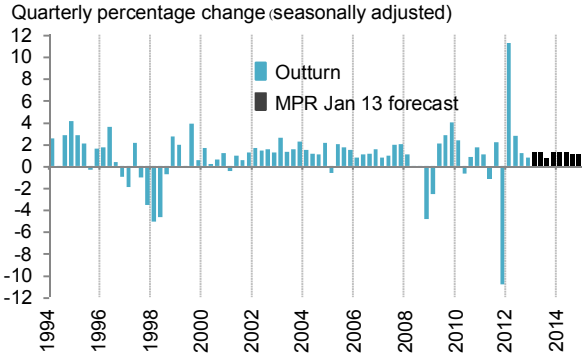
The MPC assessed that given uncertain global economic conditions and inflation outlook remaining in the target range, maintaining the policy rate at the current level was appropriate in supporting domestic demand to sustain growth momentum going forward. The MPC thus voted to hold the policy rate at 2.75 percent per annum in its last two meetings.

1.1 Growth and inflation prospects

The Thai economy was likely to expand continuously, while inflation pressure was poised to stabilize.

The Thai economy grew at a higher-than-expected rate in the second half of 2012 on the back of robust domestic demand, despite the greater impact of the global economic slowdown on exports compared to the previous period. Meanwhile, private consumption continued to expand, supported by government stimulus measures, while business sector investments to replace and restore damages caused by the flood last year were higher than assessed. On this front, domestic demand momentum was the key factor responsible for the higher-than-assessed

Chart 1.1 Thailand's GDP growth



Source: Office of the National Economic and Social Development Board and calculation by Bank of Thailand

Note: At 1988 prices (seasonally adjusted)

economic expansion in 2012. Looking ahead, the MPC evaluated that the Thai economy was likely to experience strong continual growth from domestic spending, thanks to additional support compared to the previous projection. In particular, the first-car tax rebate scheme, a personal income tax cut, as well as investments both for ongoing repairs and replacements in some industries and to accommodate strong domestic demand. As a result, Thai economic growth projection for 2013 was slightly adjusted upwards compared to the previous projection. The MPC assessed that exports would recover gradually and regain its role as a driver of growth in the latter half of 2013. **(Chart 1.1 and Table 1.1)**

Table 1.1 Forecast summary

Percent	2012	2013	2014
GDP growth	5.9	4.9	4.8
	(5.7)	(4.6)	
Headline inflation	3.0*	2.8	2.6
	(3.0)	(2.8)	
Core inflation	2.1*	1.7	1.6
	(2.1)	(1.7)	

Note: * Outturn
( ) Inflation Report October 2012
Source : Office of the National Economic and Social Development Board and calculation by Bank of Thailand

The major downside risk to growth was the fragile global economy, particularly in developed economies which were likely to encounter delays in recovery. Nevertheless, risks from the overall global economy were lower than previously assessed, due to progress in the resolution of Greece's sovereign debt problem which helped lower the risk of Greece leaving the euro area. In addition, uncertainty surrounding the US fiscal cliff declined after the House of Representatives and the Senate reached an agreement. Domestic risk was higher than

previously anticipated, but only slightly. Meanwhile, public sector risks could surface from uncertainty and delays in some investment projects.

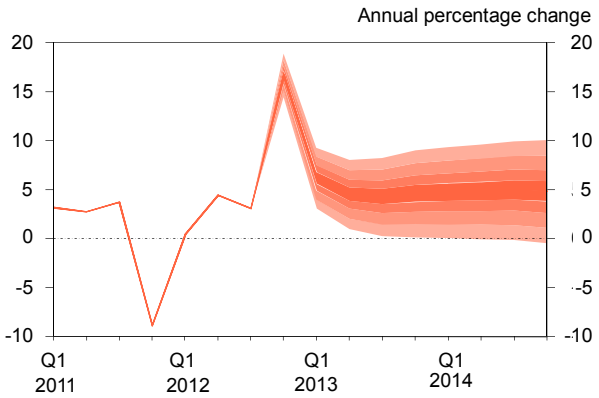
Overall, the MPC assessed that downside risks to growth, mainly from global economic headwinds, would dominate upside risks. Accordingly, the fan chart for growth remained skewed downward but to a lesser degree compared to the previous projection **(Chart 1.2)**.

Inflation pressure was expected to remain subdued, similar to the previous projection. Cost pressures subsided somewhat, in line with the price trend of domestic fresh food and global non-fuel commodities. Meanwhile, demand pressure was likely to increase on account of stronger momentum from domestic spending. The MPC thus kept the previous projection for inflation in 2013 and assessed that inflation would remain stable in 2014 (Table 1.1). Meanwhile, core inflation would remain around the mid-point of the target range. Going forward, risks to inflation were expected to balance out, compared to the previous assessment where risks were weighed to the downside. This was because downside risks from the world economic trend subsided from the previous assessment. The fan charts for inflation were thus balanced over the projection period, compared to those previously skewed downward **(Charts 1.3 and 1.4)**.

1.2 Economic outlook

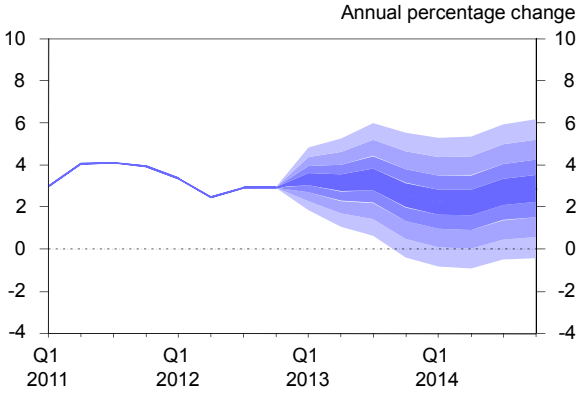
Momentum from robust domestic demand should continue, while the export sector would gradually recover and lend more support to growth in the periods ahead.

Chart 1.2 GDP growth forecast



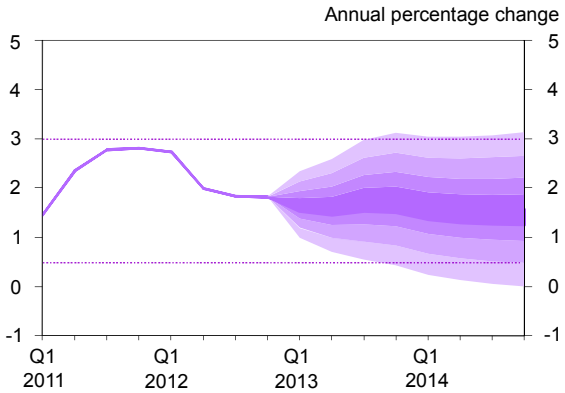
Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.3 Headline inflation forecast



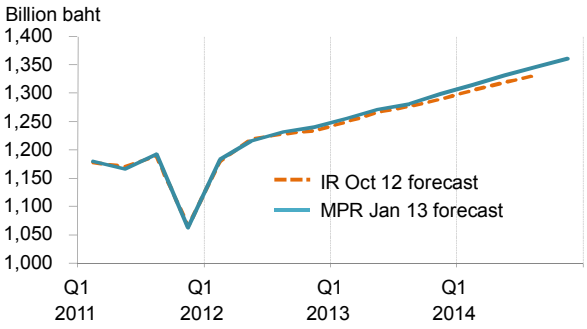
Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.4 Core inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.5 Level of GDP



Source: Office of the National Economic and Social Development Board and calculation by Bank of Thailand

Note: At 1988 prices (seasonally adjusted)

The Thai economy was expected to expand continuously.

In the latter half of 2012, the Thai economy expanded well following robust private demand, despite the growing impact of the global economic slowdown on exports. Momentum from private spending continued and was higher than previously assessed. This was partly due to more-than-expected repair and replacement investments, especially in the integrated circuit industry. In addition, some businesses resumed planned investments after they ascertained that flood problems would not recur at the end of the year. Private consumption was also supported by the first-car tax rebate scheme, with the number of subscriptions exceeding expectation. In 2012, private spending, which had not been significantly affected by the slowdown in exports, was the main contributor to the slightly higher-than-projected economic growth **(Table 1.1)**.

Looking ahead, the MPC evaluated that Thai economic growth was likely to be higher than previously assessed **(Table 1.1 and Chart 1.5)**. Domestic demand would continue to serve as the main driver of growth in 2013, with private consumption fueled by momentum from the first-car tax rebate scheme and a cut in personal income tax in the 2013 tax year. Moreover, private investment was likely to expand robustly following an increase in investments to accommodate strong domestic demand and ongoing reconstruction in some industries. Meanwhile, the export sector was expected to recover gradually before resuming its role as engine of economic growth in the latter half of 2013, which would help sustain growth after effects from government stimulus measures taper off **(Table 1.2)**.

Table 1.2 Forecasts for GDP and components			
Percent	2012	2013	2014
GDP growth	5.9	4.9	4.8
Domestic demand	8.4	6.5	5.4
Private consumption	6.3	4.3	3.5
Private investment	16.5	12.1	9.1
Government consumption	6.0	2.2	6.9
Public investment	7.9	17.1	7.3
Exports of goods and services	3.1	8.0	8.7
Imports of goods and services	6.7	10.7	10.3

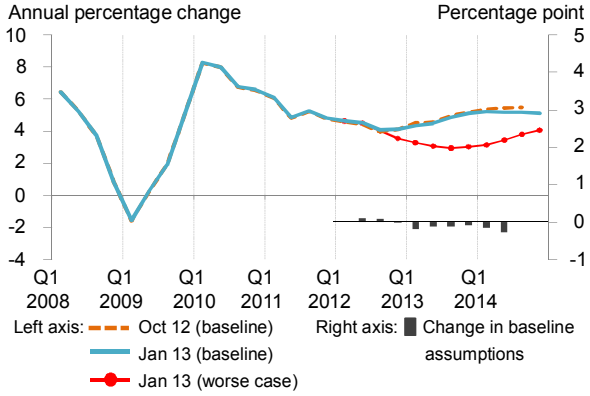
Note: At 1988 prices

The global economy showed signs of improvement.

Although the overall global economy was likely to remain weak, key economic data recently reflected steady improvements in economic stability.

Thailand's trading partners' economies were expected to recover gradually in 2013 **(Chart 1.6)**. Recovery of the US economy as well as growth in China and emerging economies would help drive the global economy, particularly in the second half of the year. However, economies of the euro area and Japan were likely to weaken from the previous assessment. Recovery in the overall global economy was in line with the direction of key economic indicators, such as steady upward adjustments of the Global Purchasing Manager Index (PMI) in the past periods.

Chart 1.6 Growth assumptions for Thailand's trading partners



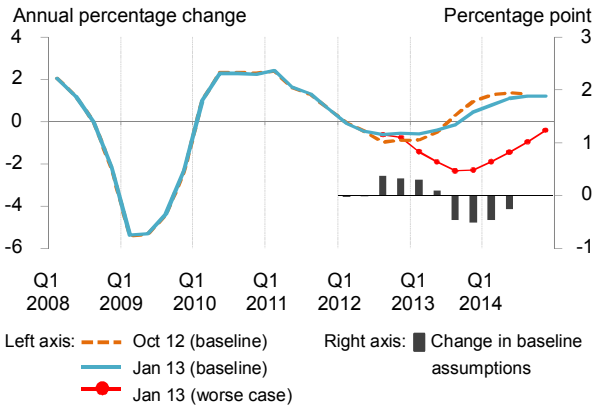
Note: Weighted by each country's share in Thailand's total exports

The euro area's economy was temporarily supported by short-tem improvements in confidence **(Chart 1.7)**. This was due to progress in the resolution of Greece's sovereign debt crisis and the announcement of the European Central Bank (ECB)'s Outright Monetary Transactions (OMTs) in the previous periods. The MPC assessed that euro area economies would recover slowly, mainly bolstered by Germany's solid economic fundamentals,

an increase in non-euro exports following recovery in global demand, as well as improving signals from the manufacturing sector. Nonetheless, the recovery trend was expected to soften from the previous assessment. This was partly due to implementation of fiscal consolidation by many countries which would deter the real economy and delay recovery. Consequently, euro area economies in 2013 were expected to contract from the previous year, with fiscal austerity measures and financial institutions' strict credit conditions constraining domestic demand recovery going forward.

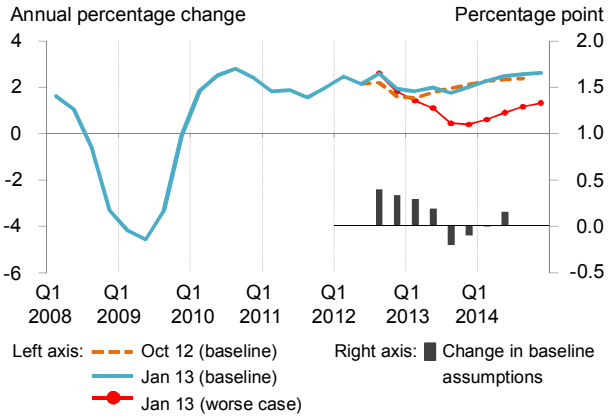
On this front, the MPC viewed that downside risks from the euro area declined markedly from the previous assessment, thanks to the lower risk of Greece exiting the euro area. This was a result of the government's success in restructuring debt in the previous periods, as well as additional assistance from the euro area, namely, an agreement to relax loan terms and an extension of the fiscal consolidation timeframe. This reduction in risk was the main reason for the significant upward adjustment on the euro area's economic growth assumption in the worse case scenario **(Chart 1.7)**.

Chart 1.7 Growth assumptions for the euro area



The US economy was expected to recover gradually over the projection period **(Chart 1.8)**. Although private consumption was boosted by a pickup in the housing sector and labor market, overall demand would remain weak for some time. Nevertheless, uncertainty regarding the fiscal cliff declined after the House of Representatives and Senate came to an agreement on extending several fiscal stimulus measures which were to be terminated end-2012. These included a cut in the personal income tax rate and an extension of federal unemployment benefits, which would help shore up private spending, as well as help reduce downside risks to US economic growth going forward.

Chart 1.8 Growth assumptions for the US



Nonetheless, uncertainties arising from debt ceiling and sequestration, which need to be finalized by February 2013, remained key risks to the economy and could affect private sector sentiment in the periods ahead.

Asian economies were poised to pick up in 2013 on account of strong domestic demand which would help ease impacts from the export sector **(Chart 1.9)**. The manufacturing sector became more stable, as reflected by a rise in many countries' Manufacturing Purchasing Manager Index **(details in Chapter 2, Chart 2.5)**. Furthermore, signs of improvement were also found in export demand, as reflected by key indicators, such as the US' ISM New Orders which started to improve in line with economic recovery in the US, one of Asia's major export markets. Meanwhile, the Chinese economy was driven mainly by domestic demand, with consumption expecting to lend more support to growth later on. Compared to the previous projection, the MPC assessed that risks from Asian economies would increase from Japan's weakening economic trend **(Chart 1.10)**, after Japan's exports declined markedly in the second half of 2012. As a consequence, private investment conditions and sentiment were also affected. An additional risk could also arise from prolonged territorial dispute between China and Japan.

Chart 1.9 Growth assumptions for Asian economies

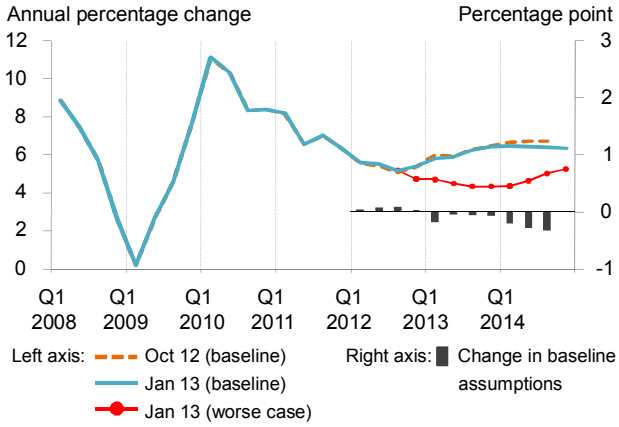
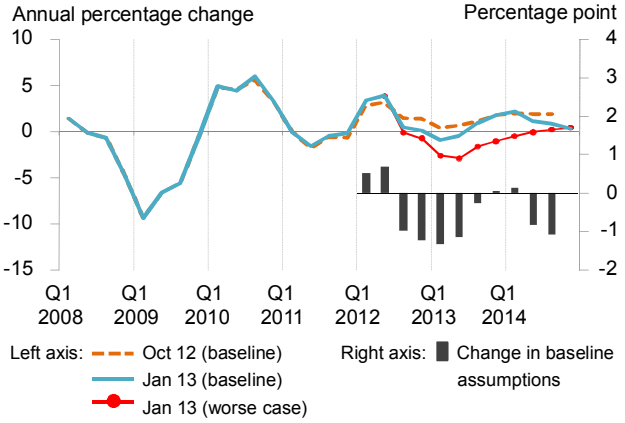


Chart 1.10 Growth assumptions for Japan



Thailand's export sector would recover gradually in line with the global economy.

Thai exports started to recover and would have a bigger role in the Thai economy in the second half of 2013. Meanwhile, imports were likely to rise in line with a pickup in domestic demand and exports.

Merchandise exports were still hindered by the global economy, but should have passed their trough, with signs of gradual improvement **(Table 1.3)**. Exports of manufacturing products started to show signs of stabilization, with a more concrete

recovery expected in the second half of 2013. This was in line with unraveling of constraint problems in IC production, and a gradual pickup in world demand. Meanwhile, exports of agricultural products in 2013 were likely to stabilize before increasing slightly. Rice exports were expected to expand from the previous year which experienced lower-than-normal exports. In addition, exports of rubber were expected to benefit from China's economy which could recover faster than assessed. The gradual recovery in Thai exports was consistent with the region's export trend which also started to display signs of recovery. Meanwhile, the tourism sector was expected to expand continuously.

Table 1.3 Forecasts for the external sector

	2012	2013	2014
Growth in value of exports* (F.O.B., percent)	3.6	9.0	9.7
Growth in value of imports* (F.O.B., percent)	8.3	9.2	9.3
Trade balance (billion U.S. dollars)*	8.1	8.4	10.1
Current account balance (billion U.S. dollars)*	2.2	1.2	0.9

Note: *Data revision according to definitions in IMF's Balance of Payments Manual, 6th edition (BPM6), and Ministry of Commerce's revised database

The MPC assessed that the export sector would recover gradually before taking on a greater role in driving the Thai economy in the latter half of 2013 and into 2014, alongside private demand and public investment. Nevertheless, risks to export recovery in early 2013 remained, as global demand for some products had not yet shown clear signs of recovery, particularly in electronics. This was reflected in key indicators, such as the book-to-bill ratio in the US' semi-conductor industry, which indicated that demand was weak and inventory level remained high.

Merchandise imports were expected to decelerate back to its normal trend in 2013 after high acceleration in the previous year. Imports of consumer goods were expected to expand well in line with robust domestic demand. Meanwhile, imports of capital goods were likely to decelerate from the previous period, following investments in flood reconstruction which would taper off soon. The MPC viewed that quarterly momentum would decline from end-2012 which saw more-than-normal acceleration due to temporary factors. These factors included the surge in imports of machinery and equipment once businesses were certain that flood problems would not recur at the end of the year, as well as acceleration in imports before exemption of import duty on machineries to replace those damaged by the flood expired in end-2012.

Relative to the previous assessment, the MPC expected the current account balance[1] to show a lower surplus in 2013 **(Table 1.3)**, owing to higher merchandise imports in line with domestic demand. For 2014, the current account surplus was expected to decline from the previous year, partly due to claim payouts from foreign insurance companies which were expected to end.

Continual momentum from private demand

Private consumption and investment would continue to expand well, backed by many strong supporting factors.

Private consumption would continue to experience above-normal expansion for some time and would remain the main engine of economic growth over the projection period. This was boosted by favorable household income, as well as continuous government stimulus measures, particularly from the first-car tax rebate scheme with tax return payments beginning in October 2012. At the same time, the

[1] Including reinvested earnings

increase in automobile expenditures did not lower spending on other consumer goods. Compared to the previous projection, the MPC evaluated that private consumption in 2013 would be higher than previously projected, before slowing down slightly in 2014 given high consumption in the previous periods, particularly in automobiles. Domestic supporting factors increased from the previous projection, fueled by more-than-expected subscriptions for the first-car tax rebate scheme as well as the ability of most producers to expedite delivery of cars. In addition, cutting the personal income tax rate for the 2013 tax year would increase after tax income, which was expected to boost household spending to some extent.

Looking ahead, many factors supporting consumption remained strong and would help sustain consumption momentum after some government stimulus measures started to taper off in the second half of 2013. Key supportive factors included (1) favorable household income prospects, given the increase in minimum wage and civil servant salary, ongoing measures to support agricultural prices, along with positive farm income prospects from improved agricultural prices--in particular the price of rubber which was expected to increase from the current level following expansion in the Chinese economy and rice prices which would remain high due to an extension of the rice pledging scheme; (2) consumer confidence remained healthy, stemming from government measures to help raise short-term purchasing power, as well as a stable inflation trend; and (3) monetary conditions remained accommodative and supportive of household credit growth, while household's ability to service debt remained satisfactory. High consumer loans, however, was a risk factor that warranted monitoring going forward.

Private investment was expected to accelerate more than previously assessed. In 2013, investment was expected to grow close to the long-term average, due to investments by businesses to accommodate domestic demand growth, particularly in automobile and power generators. Meanwhile, investments in machinery and equipment, as well as in construction were still sustained by reconstruction in some industries, such as the integrated circuit industry, which was expected to continue into early 2013. The MPC assessed that overall investment would still benefit from the up-cycle in investment which started since 2009. Furthermore, there were several supporting factors over the medium- and long-term: (1) Businesses remained confident in the Thai economy and were likely to invest according to their long-term plans. There continued to be demand for investment in Thailand, as reflected by rising values of application and issuance of investment promotion certificates by the BOI and high levels of foreign direct investment in various industries. (2) Overall monetary conditions remained supportive, with a broad-based corporate loan expansion, due partly to low interest rates and relaxed credit standards. (3) Long-term public investments in infrastructure were more concrete and would crowd in private investment going forward. And (4) investment in 3G projects, which was likely to begin in 2013.

Some key challenges to private investment needed monitoring, in particular delays in public investment on long-term water management and infrastructure projects. In addition, fragile recovery in industrial countries could affect investment decisions in export-oriented industries. With regard to the nationwide minimum wage increase to 300 baht in 2013, the MPC evaluated that it would not have a significant effect on the overall economy as most businesses had sufficient profits to cushion the impact, and had, to some extent, already adjusted by

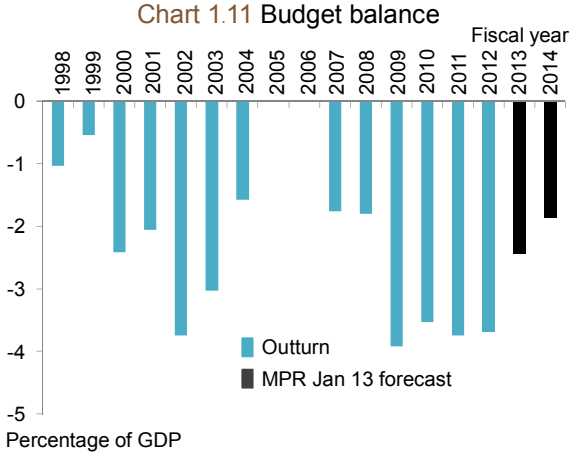
reducing expenditures and increasing productivity. Furthermore, the wage increase was not expected to affect overall employment conditions much, due to the tight labor market. At the same time, the MPC evaluated that the hike in minimum wage would also benefit the overall economy, via consumption spending and investment in the manufacturing sector to replace manual labor.

Public spending continued to provide economic stimulus

Fiscal stimulus continued at a level close to the previous projection and would play a key role in supporting private spending going forward.

Fiscal spending would continue to provide stimulus over the projection period, as reflected by a high proportion of budget deficit compared to GDP. The budget deficit for fiscal years 2013 and 2014 was expected to stand at 300 and 250 billion baht, respectively **(Chart 1.11)**. The MPC evaluated that public spending in fiscal year 2013 was likely to be similar to that in the previous projection **(Table 1.4)**. Meanwhile, consumption spending was likely to increase slightly due to a reduction in transfer payments consistent with actual data, while public investment was expected to slow down on the back of lower carry-over due to higher-than-projected investment in the previous fiscal year. Disbursements of the expenditure budget and investment budget in fiscal year 2013 Q1 accelerated more than expected due to back payment of the academic standing allowance and state enterprise's import of aircrafts. As a result, spending for the rest of the fiscal year decelerated slightly from the previous assumption.

Chart 1.11 Budget balance



Source: Bureau of the Budget and calculation by Bank of Thailand

Table 1.4 Assumptions on public sector expenditure

Unit: Billion baht	As of Oct 2012		As of Jan 2013	
	FY 2013	FY 2014	FY 2013	FY 2014
General government consumption	1,570.4	1,723.9	1,575.1	1,723.9
Public investment	763.6	825.0	750.8	825.0
Total	2,334.0	2,548.9	2,325.9	2,548.9

Note: FY denotes fiscal year.

Apart from the above budget expenditure, the MPC assessed that investment expenditure on the water management project of 350 billion baht could delay from that previously assessed. This would disperse disbursements out to later fiscal years. Meanwhile, long-term infrastructure investments became more concrete. Most recently, the Budget Act draft was approved by the Strategic Committee for Reconstruction and Future Development (SCRF) with investments expected to commence at the end of fiscal year 2014. The MPC evaluated that the aforementioned investment project was a key factor that would help mitigate limitations of the manufacturing sector and build up business confidence in the long-term.

1.3 Inflation pressure

Inflation pressure was likely to remain stable, similar to the previous assessment. The slightly rising trend of demand pressure was offset by softening cost pressure. Risks to inflation were balanced.

Inflation pressure was likely to remain stable

Overall inflation pressure stabilized, similar to the previous assessment. Demand pressure was likely to rise slightly, in line with higher-than-expected momentum from private spending. This was due to a boost in consumption from the first-car tax rebate scheme which was higher than expected and ongoing investments for reconstruction in some industries. The MPC assessed that demand pressure would continue in the periods ahead, owing to the likelihood of the Thai economy growing near its potential, as reflected by the near zero output gap over the projection period **(Chart 1.12)**.

Chart 1.12 Output Gap

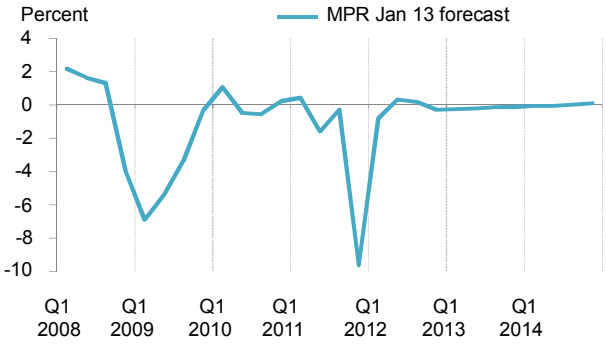


Chart 1.13 Assumptions on fresh food prices

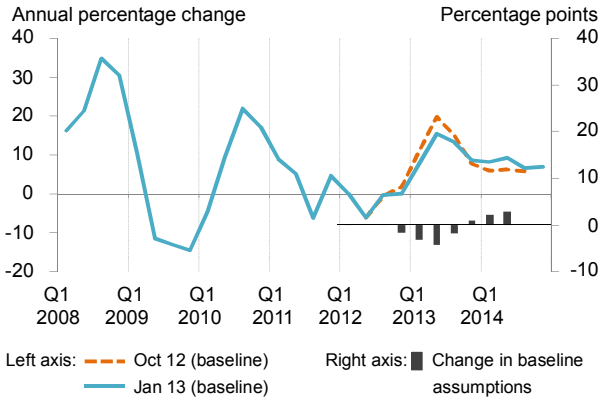


Chart 1.14 Assumptions on non-fuel commodity prices

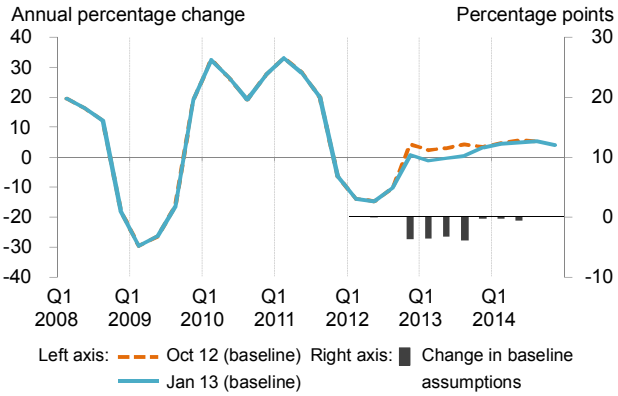
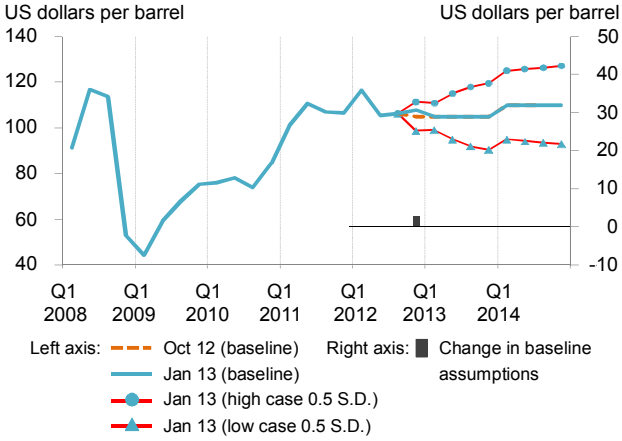


Chart 1.15 Assumptions on Dubai oil price



Cost pressure, however, was likely to soften somewhat, partly due to lower expansion in domestic fresh food prices in 2013 **(Chart 1.13)**, following higher-than-expected supply of meats into the market. Furthermore, the pass-through of higher global grain prices, from drought problems in the US, to costs of animal feed and prices of pork, poultry and eggs was relatively low. Meanwhile, rice prices were likely to remain stable. In addition, non-fuel commodity prices in the global market were also expected to soften from the previous assessment **(Chart 1.14)** given abundant supply of food products than previously expected due to an increase in planting area and yield. Prices in 2013 were expected to remain stable before rising gradually in line with world demand which was expected to recover more clearly in the latter half of the year.

Global crude oil prices were likely to remain stable over the projection period **(Chart 1.15)**. The MPC projected baseline assumptions on Dubai oil price in 2013 and 2014 at 105 and 110 US dollars per barrel, respectively, unchanged from the previous assessment. High case and low case assumptions on Dubai oil price remained balanced at half a standard deviation above and below the baseline path, because demand and supply fundamentals were assessed to be somewhat balanced and similar to the previous assessment. Crude oil production in non-OPEC countries was likely to increase following resumed production after the shutdown for maintenance ended. Nevertheless, OPEC was expected to manage overall oil production to balance with world demand in order to maintain oil prices close to the current level.

In this projection, the MPC adjusted risks to inflation projection to be balanced, from that skewed downward in the previous projection. This was because downside risks from the domestic economy were likely to subside following strong momentum of

private spending and limited adverse effects from the export sector. Meanwhile, risks from the global economy also receded following clearer actions regarding Greece's sovereign debt problem and US' fiscal problem in the past periods. Still, some upside risks to inflation remained from tensions in the Middle East which could pressure global oil prices, and from the impact of the minimum wage hike on domestic prices which could be greater than estimated.

1.4 Monetary policy decision

Policy stance remained accommodative

The MPC deemed the current policy rate appropriate in supporting domestic demand to sustain Thai economic growth. The MPC thus voted to hold the policy rate in its last two meetings.

In its meeting on November 28, 2012, the MPC judged that the global economy remained stable and exhibited signs of improvement. US economic data were better than expected, especially in the labor and housing markets, although the fiscal cliff remained a key risk factor. Meanwhile, Chinese and Asian economies also showed signs of improvement. Thailand was poised to expand continuously, with global impact limited only to export-related sectors. Exports were expected to begin recovery in the first half of 2013, on the back of global economic recovery. Meanwhile, private consumption and investment would continue to be the main drivers of economic growth going forward. Inflation pressure stabilized at a moderate level close to the previous meeting. The MPC viewed that as downside risks to growth subsided and with inflation pressure in check, the current policy rate remained accommodative and conducive to growth. The MPC therefore voted unanimously to maintain the policy

rate at 2.75 percent per annum and would stand ready to take appropriate policy action as warranted.

In its subsequent meeting on January 9, 2013, the MPC assessed that the overall world economy recovered gradually, with improving signals from the previous meeting, led by the US and China. In addition, the recent agreement to avert the US fiscal cliff helped boost global financial market sentiment. Meanwhile, the Thai economy in 2012 Q4 was likely to expand more than previously assessed. Consequently, economic growth in 2012 and growth projection for 2013 were expected to be higher than previously projected, driven mainly by strong private consumption and investment. Furthermore, the export sector showed incipient signs of a broad-based recovery, while the services sector and tourism expanded robustly. Inflation pressure remained stable, close to the previous meeting. Nevertheless, the impact of a second-round minimum wage increase warranted monitoring. The MPC assessed that the accommodative monetary policy stance throughout the previous year had significantly shored up private sector confidence, supported post-flood recovery, and helped cushion the economy from the global economic headwinds. The MPC viewed that, with remaining uncertainties in the global economy and inflation forecast within target, the current monetary policy stance was appropriate in supporting domestic demand to sustain growth momentum. The MPC thus voted unanimously to maintain the policy rate at 2.75 percent per annum. However, the MPC would closely monitor financial stability risks that could arise from persistently high credit growth, rising household debt, and volatile capital flows.

1.5 Appendix:

Table 1.5 Forecast assumptions			
	2012	2013	2014
Dubai oil price (U.S. dollars per barrel)	109.0	105.0	110.0
Non-fuel commodity prices (%YoY)	-9.9	0.6	4.7
Fresh food prices (%YoY)	-1.5	11.2	7.7
Minimum wage in the Bangkok Metropolitan Region (baht per day)	279	300	300
Government consumption (%YoY) 1/	8.8	6.8	9.8
Public investment (%YoY) 1/	11.8	23.9	13.1
Fed Funds rate (% at year-end)	0.13	0.13	0.13
Trading partners' economic growth (%YoY) 2/	4.3	4.7	5.2
Regional currencies vis-à-vis the U.S. dollar (Index) 3/	108.5	105.7	104.0

Note: 1/ Including spending on water management plans
2/ Weighted by each country's share in Thailand's total exports
3/ Appreciation against the U.S. dollar indicated by a decrease

Table 1.6 GDP growth forecasts by research houses		
	2013	2014
Kasikorn Research	5.0	4.9
DBS	5.0	n.a.
NESDB 1/	4.5-5.5	n.a.
FPO 2/	4.5-5.5	n.a.
BOT	4.9	4.8
TMB	4.7	6.1
HSBC	4.6	4.1
SCB EIC	4.6	n.a.
Phatra Securities	4.5	5.0
UBS	4.5	4.5
JP Morgan Chase	4.5	n.a.
Tisco Securities	4.5	n.a.
UOB	4.0	n.a.

Note: Compiled and published by Reuters on January 11, 2013, except:
1/ Published on November 19, 2012, with the release of GDP data for 2012 Q3
2/ Published on December 26, 2012
Presented in descending order of 2013's forecast

Table 1.7 Headline inflation forecasts by research houses		
	2013	2014
UOB	3.8	n.a.
JP Morgan Chase	3.7	n.a.
DBS	3.6	n.a.
UBS	3.5	3.1
Kasikorn Research	3.3	3.0
Tisco Securities	3.3	n.a.
HSBC	3.2	3.8
TMB	3.2	3.2
Phatra Securities	3.2	3.0
SCB EIC	3.0	n.a.
NESDB [1/]	2.5-3.5	n.a.
FPO [2/]	2.5-3.5	n.a.
BOT	2.8	2.6

Note: Compiled and published by Reuters on January 11, 2013, except:
[1/] Published on November 19, 2012, with the release of GDP data for 2012 Q3
[2/] Published on December 26, 2012
Presented in descending order of 2013's forecast

Table 1.8 Probability distribution of GDP growth forecast									
Percent	2012	2013				2014			
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 24	0	0	0	0	0	0	0	0	0
21-24	0	0	0	0	0	0	0	0	0
18-21	16	0	0	0	0	0	0	0	0
15-18	73	0	0	0	0	0	0	0	0
12-15	11	0	0	0	0	0	1	1	1
9-12	0	6	2	2	5	6	7	9	8
6-9	0	46	23	19	26	25	26	28	26
3-6	0	42	51	46	42	40	39	37	36
0-3	0	5	22	28	23	23	23	20	22
(-3)-0	0	0	2	5	4	5	5	5	6
< (-3)	0	0	0	0	0	0	0	0	1

Table 1.9 Probability distribution of headline inflation forecast								
Percent	2013				2014			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 7	0	0	0	0	1	1	1	2
6-7	0	1	1	2	2	2	3	4
5-6	3	6	5	5	5	6	8	8
4-5	19	18	12	11	10	12	14	14
3-4	41	30	21	18	17	18	19	19
2-3	30	27	24	22	21	21	20	19
1-2	7	14	19	20	20	18	17	16
0-1	1	4	11	13	14	12	11	10
(-1)-0	0	1	4	7	7	6	5	5
< (-1)	0	0	1	3	4	3	3	3

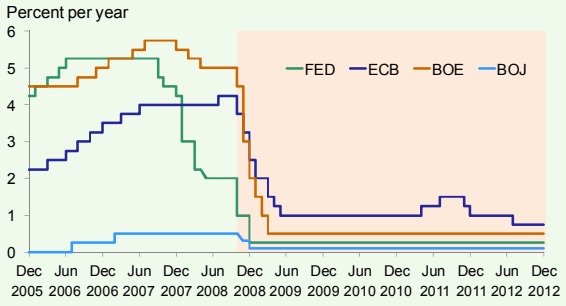
Table 1.10 Probability distribution of core inflation forecast								
Percent	2013				2014			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 4.0	0	0	0	0	0	0	0	0
3.5-4.0	0	0	1	1	1	1	1	2
3.0-3.5	0	1	4	5	4	4	4	5
2.5-3.0	2	6	11	12	10	10	10	10
2.0-2.5	19	20	22	20	18	17	16	16
1.5-2.0	45	33	26	24	23	22	21	20
1.0-1.5	29	27	21	20	21	21	20	20
0.5-1.0	5	11	11	11	13	14	15	14
0.0-0.5	0	2	3	4	6	7	8	8
< 0.0	0	0	1	2	3	4	4	5

Challenges from monetary easing by major economies and implications on Thailand

Recent monetary policy implementation by major economies

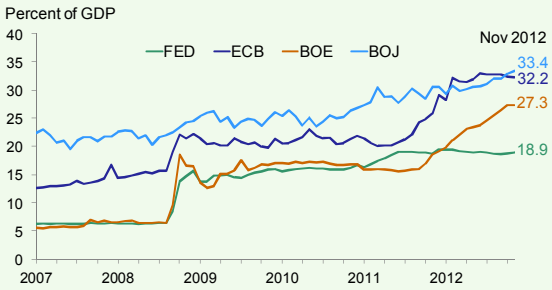
During the past 4-5 years, the global economic crisis and public debt problem in the euro area have caused prolonged recessions in major economies, particularly the US, the euro area, the UK and Japan. Despite many years of low or near-zero policy rates **(Chart 1)**, not much positive economic impact has materialized partly owing to low confidence of the private sector and investors coupled with uncertain income prospects from weak labor market conditions, leading consumers to slow spending, thus creating a drag on recovery.

Chart 1 Short-term interest rates of major economies



Note: The Federal Reserve (FED) The European Central Bank (ECB) Bank of England (BOE) and Bank of Japan (BOJ)
Source: Bloomberg

Chart 2 Assets of major economies' central banks
(percent of GDP)



Note: The Federal Reserve (FED), The European Central Bank (ECB), Bank of England (BOE) and Bank of Japan (BOJ)
Sources: FED, ECB, BOE, BOJ, and CEIC

Against the backdrop of longer-than-expected recovery of major economies and zero lower bound limitation on policy rates, many central banks have opted for unconventional measures such as Quantitative Easing (QE) in the US [1/], Outright Monetary Transactions (OMTs) in Europe[2/], and Asset Purchase Programs in Japan and the UK. These measures have caused the size of central banks' balance sheets to expand **(Chart 2)**, but the direct economic impact has still been small due to overall weaknesses in financial conditions, leading financial institutions to hold extra reserves instead of giving out loans.

In addition, there has been a major change in central banks' communication. For example, the Federal Reserve and Bank of Canada have been communicating with forward guidance in order to lower long-term interest rates in the financial markets. Most recently, the Federal Reserve has communicated its policy rate path with reference to a macroeconomic target, announcing to keep the policy rate at 0-0.25 percent per annum given that the unemployment rate remained above 6.5 percent and inflation projection between 1-2 years ahead was no more than half a percentage point above the Committee's long-term inflation target of 2 percent. This new communication strategy contrasts with the previous one where referencing has been made to calendar years, for example, an announcement to maintain the policy rate until mid-2015.

Despite implementations of both conventional and unconventional measures by major economies' central banks, their economic performance has remained weak and unemployment

---

[1/] QE refers to the purchase of long-term government bonds and mortgage-backed securities (MBS) by the Federal Reserve to support recovery in the US labor and real estate markets. Four QEs have been announced since November 2008 with the latest announcement (QE3+) in December 2012.

[2/] OMTs refer to the unlimited purchase of Member States' government bonds in the secondary markets by the ECB upon the requests of Member States.

has been high. As a result, it has been suggested that these economies move to target nominal GDP (NGDP level targeting) whereby a nominal GDP growth target could be set to equal the long-term average of real GDP growth plus the central banks' inflation target in order to further accommodate economic recoveries.

The concept of NGDP level targeting is "history dependence". Suppose that the NGDP target is 4.5 percent per year. If actual NGDP growth is less than 4.5 percent in a particular year, i.e. lower than the target path rate, accommodative monetary policy will be required to compensate for the missed target by stimulating economic growth beyond 4.5 percent until the NGDP returns to its target path. This type of monetary policy where bygones are not bygones is different from the forward-looking kind. For example, in inflation targeting, a policy stance is deemed appropriate if it can achieve a balance between growth and inflation in future periods regardless of current inflation and growth outturns.

Nevertheless, NGDP level targeting is still a theoretical suggestion at this stage given many practical limitations. (1) Determination of long-term average of real GDP growth is difficult. (2) The target lacks clarity as it is a combination of GDP and inflation. This problem makes communication with the public difficult, especially in explaining the central bank's inflation target, which could ultimately un-anchor inflation expectations and affect the private sector's assessment of future financing costs. (3) Policy implementation lacks flexibility given the low frequency (quarterly) and long lag in NGDP announcement as well as data revisions.

Implications to Thailand's policy implementation

Looking ahead, it is expected that major economies would remain weak for a while. Thus, even the use of NGDP level targeting would not guarantee a faster turnaround. On the other hand, central banks of major economies would need to conduct monetary easing for much longer to compensate for the missed target path since the global crisis. This would, in the end, create challenges to policy implementation of emerging economies, including Thailand, as follows.

1. Implication on monetary policy in Thailand. Recent weaknesses in major economies have caused Thai exports to slow down during some periods, posing risk to overall economic performance, going forward. Should major economies continue to fail to stimulate growth, greater risk would fall on the Thai economy and ultimately affect monetary policy stance.

2. Implication on foreign exchange and capital flows. Major economies' near-zero policy rates have caused interest rate differentials between these economies and emerging economies (including Thailand) to widen. Moreover, continuous liquidity injections have also resulted in searches for yields in emerging economies, including Thailand, where growth prospects are promising. Such developments could have significant impacts on volatility of capital flows and the Thai baht.

Nevertheless, the MPC has assessed that Thailand's flexible inflation targeting (FIT) framework is appropriate to the country's economic conditions and structure, particularly from its transparency and ease of communication. Moreover, many countries that have also been using this framework also register satisfactory growth performance. In Thailand's case, since the adoption of FIT in 2000, growth performance has been satisfactory. Meanwhile, economic stability has been well maintained, as reflected by low inflation and absence of imbalances. The framework has also been useful in cushioning the impact from the recent global economic crisis. [3/]

[3/] IMF's latest research (WP/12/269) noted that without FIT and flexible exchange rates, the Thai economy would have been more severely affected by the global economic crisis. For example, if Thailand had used the fixed exchange rate policy, growth would have been 3.7 percent lower than actual outturn, which is considered quite high considering the normal growth rate of around 5 percent per year.

Recent Economic Developments

2. Recent Economic Developments

The world economy as a whole recovered gradually, showing signs of improvement, particular in the US and China. Moreover, the recent agreement to avert the US fiscal cliff, as well as the euro area's financial assistance and additional supportive measures for Greece helped bolster global financial market sentiment. Nonetheless, the euro area and Japanese economies remained weak, while resolution of their structural problems would likely take time. Meanwhile, economic performance of most Asian economies turned more positive on the back of strong domestic demand as well as modestly improving exports.

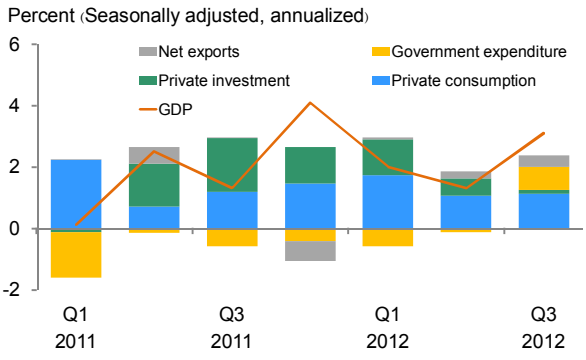
The Thai economy in 2012 Q4 was expected to expand from the previous quarter following strong domestic demand, supported by consumer and business confidence, favourable household income and high employment, as well as accommodative monetary conditions. Meanwhile, the export sector showed incipient signs of a broad-based recovery. Inflation pressure should remain moderate going forward. Nevertheless, the impact of the second-round minimum wage hike warrented monitoring.

2.1 The global economy

The US economy recovered gradually following improvements in employment conditions which facilitated continued growth in private consumption. Moreover, clearer signs of recovery in the housing sector were also observed.

In 2012 Q3, the US economy grew by 3.1 percent from the previous quarter **(Chart 2.1)** due to imperative positive factors. These included a jump in private consumption growth, increased investment in the housing sector, and acceleration in government spending on national defense, the latter a temporary factor.

Chart 2.1 Contribution to US economic growth
(Change from the previous quarter)

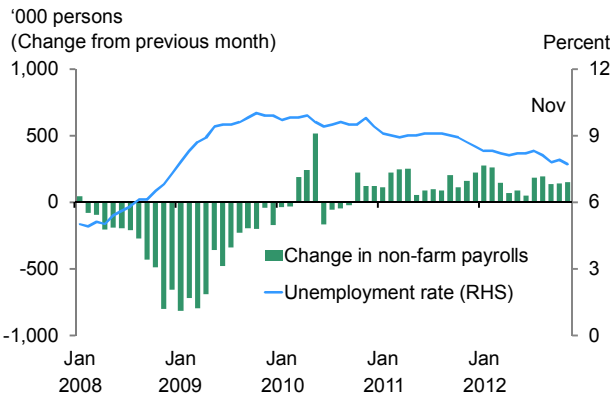


Source: Bureau of Economic Analysis

In 2012 Q4, the US economy continued to expand, due mainly to improvements in private consumption. This was reflected by a surge in retail sales, particularly in automobile sales which had remained high for several consecutive months, consistent with improving consumer confidence.

The pickup in private consumption was mainly driven by the following supporting factors. (1) Continued improvement in employment conditions **(Chart 2.2),** which led to a reduction in the unemployment rate to stand at 7.7 percent in November 2012, the lowest level since end-2008. Nonetheless, the lower unemployment rate was partly due to a decline in labor force as some of the unemployed had left the workforce. This reflected that adjustments in the labor market would take time. And (2) clearer signs of recovery in the housing sector, as reflected by continued improvement in home sales and continued increases in housing prices compared to the same period of the previous year. This increase in housing prices helped improve households' financial position and would support private consumption going forward.

Chart 2.2 US labor market conditions



Source: Bureau of Labor Statistics

Looking ahead, the MPC viewed that the US economy would recover gradually following FOMC's accommodative monetary policy stance. US growth, however, would remain subpar due to limitations of the fiscal stimulus package and the somewhat high unemployment rate. In December 2012, the Fed announced additional economic stimulus policies which included (1) the purchase of long-term Treasury securities of 45 billion US dollars a month to replace the Maturity Extension Program (Operation Twist) which completed at the end of 2012, and (2) keeping the federal funds rate at 0-0.25 percent per annum, given that the unemployment rate remained above 6.5 percent and inflation projection between 1-2 years ahead was no more than half a percentage

point above the Committee's inflation target of 2 percent. Moreover, the Fed would continue to purchase agency mortgage-backed securities (QE3) at a pace of 40 billion US dollars per month to support continuous recovery in the labor market and housing sector which had already started to show signs of recovery.

Moreover, the main factor driving economic recovery in the US going forward would be the gradual and continuous improvement in employment conditions, especially in the business sector following an increasing investment trend. Despite progress in cutting the personal income tax rate and extension of federal unemployment benefits, fiscal cliff uncertainties arising from debt ceiling and sequestration, which need to be finalized by February 2013, remained key risks to the economy and could affect short-term confidence.

The euro area's economy weakened during the latter half of 2012. However, the risk of Greece exiting the euro area subsided after Member States approved additional assistance to Greece. This was expected to support economic recovery of the euro area in 2013.

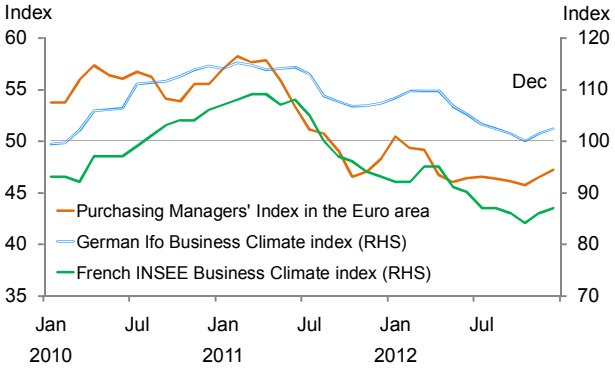
In 2012 Q3, the euro area economy contracted by 0.1 percent from the previous quarter owing to fiscal consolidation measures in almost every Member State and financial institutions' strict lending standards, as well as the slowdown in private spending due mainly to the prolonged and uncertain sovereign debt crisis.

The risk of Greece exiting the euro area subsided. In the December 13, 2012 meeting, euro area finance ministers jointly concluded on further assistance to Greece. This included extending Greece's timeframe to reduce its fiscal deficit, extending loan maturity, reducing loan interest rates, as well as approving the disbursement of financial aid to Greece

of 41.9 billion euro. Moreover, Greece was successful in buying back its bonds held by private investors of 31.9 billion euro, which helped lower sovereign debt by 9.5 percent of GDP. Due to these measures and the buy-back of government bonds, Greece's sovereign debt was expected to fall to 124 percent of GDP by 2020, which would improve fiscal sustainability.

The lower risk of Greece exiting the euro area together with the European Central Bank's (ECB) announcement to undertake Outright Monetary Transactions (OMTs) led to an increase in the Composite PMI, both the Manufacturing PMI and the Services PMI, as well as improvements in the German Ifo Business Climate index and the French INSEE Business Climate index **(Chart 2.3)**.

Chart 2.3 Purchasing Managers' Index (PMI) and Economic Sentiment Indicator



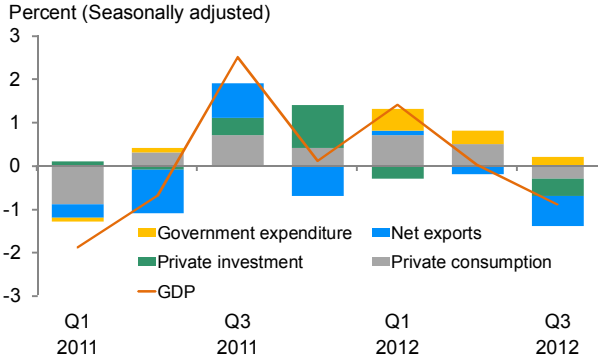
Source: Bloomberg

The MPC assessed that the euro area economy remained weak with gradual recovery expected in 2013. Despite favorable factors from stronger confidence and an increase in non-euro area export demand in light of world economic recovery, there remained risks that could delay recovery. These risks included the prolonged and uncertain sovereign debt crisis which could heighten, and higher-than-expected adverse effects of fiscal consolidation measures on the euro area economy.

The Japanese economy weakened significantly during the second half of 2012, but was expected to resume recovery in 2013, driven by recovery in trading partner economies and the new government's stimulus measures.

In 2012 Q3, the Japanese economy contracted by 0.9 percent from the previous quarter, due to contractions in both exports and domestic demand **(Chart 2.4)**. Exports declined in almost every major export market following weak global demand and appreciation of the yen. Main contracting exports included automobiles and capital goods. Moreover,

Chart 2.4 Contribution to Japan's economic growth
(Change from the previous quarter)



Source: Cabinet Office of Japan

contractions in domestic demand were on account of lower income and consumer confidence following export shrinkage and a reduction in the government's stimulus package.

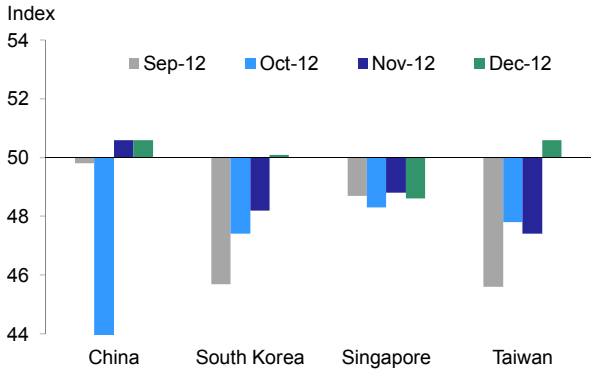
The Japanese economy was likely to contract further in 2012 Q4, with exports remaining fragile. Although exports to the US expanded, exports to China were expected to contract further. This was due to a plunge in Chinese demand for Japanese goods following the dispute between Japan and China over the Diaoyu or Senkaku Island. Meanwhile, domestic demand remained sluggish, particularly in private investment, as reflected by BOJ's Tankan Survey in 2012 Q4 on the Business Confidence Index of large industries which was revised down from the previous quarter, and was likely to fall further going forward.

The MPC assessed that Japan's economy would remain sluggish, with slow recovery in 2013, thanks to trading partners' economic recovery and the new government's stimulus package. The Japanese government, which took office in December 2012, was expected to implement accommodative monetary and fiscal policies to help stimulate the economy in 2013 Q1. Nevertheless, effects from this stimulus package could be limited and would need to be monitored further *(more details in BOX: Outlook on Japanese economic policy after the December 2012 election)*.

Chinese and many Asian economies improved in 2012 Q4 compared to the previous quarter, owing to more stable production and exports.

In 2012 Q4, most Asian economies picked up from the preceding quarter due to continual recovery in the manufacturing sector as reflected by improvements in the Purchasing Managers Index in many countries **(Chart 2.5)**. Meanwhile, the export sector also displayed signs of improvement,

Chart 2.5 Manufacturing Purchasing Managers' Index



Source: CEIC and Market Economics

particularly in South Korea and Taiwan. Nonetheless, recovery in exports was mainly limited to telecommunication and high technology products due to healthy global demand for these products. Exports of ASEAN economies remained weak due to their smaller export share of high technology products. Domestic demand of ASEAN economies, however, continued to grow strongly and somewhat helped cushion the slowdown in exports.

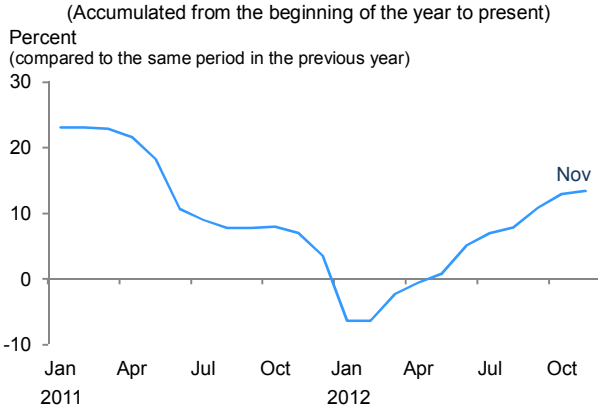
For the outlook in 2013, the MPC assessed that Asian economies would improve gradually, driven by sustained robust domestic demand. Furthermore, clearer signs of recovery in global and Chinese economies would help support recovery of regional exports going forward.

Improvements in the Chinese economy became more apparent in 2012 Q4 compared to the previous quarter. This was partly driven by the accommodative monetary policy stance and expedition of infrastructure investments in the previous periods, which started to show positive effects on economic growth in various fronts, including consumption, investment and production. Exports, meanwhile, improved in line with the more stable global economy and an increase in foreign demand for telecommunication products.

Looking ahead, the MPC assessed that the Chinese economy would improve steadily in 2013. This would be driven by domestic demand. In particular, consumption would lend more support to economic growth on account of favorable employment conditions and income level following wage increases. Other supporting factors that would help drive the Chinese economy in 2013 included expedition of public investment in both infrastructure investments **(Chart 2.6)** and investments to enhance productivity under the Twelfth Five-year National Economic and Social Development plan (2011-2015).

Chart 2.6 Infrastructure investment in China

(Accumulated from the beginning of the year to present)



Source: CEIC

Meanwhile, private investment was expected to recover gradually in line with recovery in the real estate sector, as reflected by constant increases in real estate prices and sales volume. Moreover, exports were likely to recover slowly following better world economic prospects.

2.2 The domestic economy

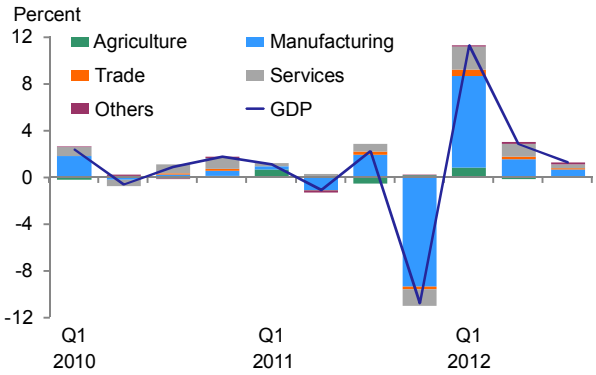
The Thai economy in 2012 Q3 continued to expand from the previous quarter, driven by strong domestic spending, from both the private and public sectors. This helped offset the slowdown in the export sector and production in export-oriented industries which were affected by the global economic crisis.

Economic conditions in 2012 Q3

In 2012 Q3, the Thai economy expanded by 1.2 percent from the previous quarter **(Chart 2.7)** due mainly to expansions in the manufacturing and services sectors. Manufacturing production rose by 1.6 percent from the preceding quarter, with growth concentrating mostly in production for domestic consumption which remained strong. In particular, automobile production grew exceptionally well due to incentives from the government's first-car tax rebate scheme. Nevertheless, the global economic slowdown caused manufacturing production in export-oriented industries to slow down significantly compared to the previous quarter, particularly, in the hard disk drive industry.

The services sector continued to expand from the previous quarter by 1.1 percent and was unaffected by the global economic slowdown. In 2012 Q3, the tourism sector grew at a satisfactory level due mainly to an increase in tourist arrivals from East Asia. Meanwhile, the number of tourists from Europe was comparable to the previous quarter, as

Chart 2.7 Contribution to GDP growth
(QoQ, seasonally adjusted)



Source: National Economic and Social Development Board

some European tourists switched to travel within Europe so they could attend the UEFA European Football Championship and the Olympic Games which was held in August. At the same time, the transportation and trade sectors expanded in tandem with strong domestic demand. Other services sectors, such as the real estate and construction sectors, also continued to grow well on the back of government mega-projects such as the mass rapid transit systems project in Bangkok Metropolitan and its vicinity.

Agricultural outputs remained high similar to that in the previous quarter, in particular production of second crop rice due to favorable water conditions and price incentives from the government's rice pledging scheme. In addition, other major agricultural outputs such as rubber, cassava and palm oil products also expanded strongly owing to favorable weather conditions, an increase in planting area and lack of plant disease outbreak.

Continual economic growth led the unemployment rate to remain low at 0.7 percent in 2012 Q3. Impacts from the minimum wage hike on April 1, 2012 on employment were not significant. Nonetheless, employment in the manufacturing sector slowed down somewhat, in tandem with manufacturing production which was affected by the global economic slowdown. Employment in the agricultural sector, on the other hand, expanded strongly in line with growth in agricultural production.

On the demand side, private consumption and investment continued to expand. This was largely attributable to government stimulus packages, accommodative monetary policy, satisfactory income and employment conditions, as well as fiscal stimulus from government spending. However, adverse effects from the global economic slowdown on Thai exports amplified.

Table 2.1 GDP growth rate						
Change from the previous quarter (seasonally adjusted, percent)	2011	2011		2012		
		Q3	Q4	Q1	Q2	Q3
GDP	0.1	2.2	-10.8	11.3	2.8	1.2
Domestic demand[1/]	1.8	1.0	-4.6	7.8	2.5	3.2
Private consumption	1.3	1.1	-3.7	6.7	0.9	2.2
Private investment	7.2	1.3	-7.6	14.4	4.3	5.4
Government consumption	1.1	2.2	-5.4	4.5	6.3	3.2
Public investment	-8.7	-3.1	-1.7	2.8	6.2	5.6
Exports of goods and services	9.5	4.6	-16.1	10.8	3.9	1.0
Imports of goods and services	13.7	7.6	-11.6	8.5	5.1	-2.3

Note: [1/] Domestic demand excluding changes in stocks

Source: National Economic and Social Development Board and calculations by Bank of Thailand

The global economic slowdown caused exports of goods and services to grow by a mere 1.0 percent from the previous quarter, compared to an expansion of 3.9 percent in 2012 Q2 **(Table 2.1)**. The decline was observed in exports of both manufacturing and agricultural products. Manufacturing products that declined markedly included electrical appliances, electronics and petroleum products. Nevertheless, exports of automobiles and metal continued to expand. Meanwhile, exports of agricultural products fell in line with rice exports due to deteriorating price competitiveness. Exports of rubber moderated in line with soft global demand. Exports of services, on the other hand, continued to improve from the previous quarter following a surge in foreign tourists, particularly from China. However, export freight and other transport receipts fell in line with lower international trade volume.

Domestic demand, both private consumption and investment, continued to grow robustly. Private consumption expanded by 2.2 percent from the previous quarter following high demand growth for durable and non-durable goods, in particular automobiles. Supporting factors included government stimulus packages, accommodative monetary policy,

as well as satisfactory income and employment conditions. Meanwhile, private investment grew by 5.4 percent from the previous quarter. Investments in machinery and equipment expanded following commercial vehicle purchases and capital goods imports. At the same time, investment in construction also increased following ongoing flood-damaged repairs and replacements.

Imports of goods and services contracted by 2.3 percent from the previous quarter. This was caused by a slowdown in exports and manufacturing production which were increasingly affected by the global economy. The contraction in imports was mainly attributable to a fall in the imports of raw material and intermediate goods, especially integrated circuits and chemicals, as well as a slowdown in the volume of imported crude oil. Meanwhile, imports of consumer and capital goods continued to expand, particularly the imports of food and beverages, as well as machinery and parts. Imports of services, however, declined from the previous quarter owing to the slowdown in transport payments following reductions in international trade transactions. Passenger payments also slowed down due to a drop in the number of Thais travelling abroad.

Fiscal stimulus would continue to maintain its role in supporting the economy as reflected by an expansion in government spending from the last quarter. In particular, wages and salaries, purchases of goods and services, as well as investment disbursements for flood restoration and prevention and for transportation system improvements.

The outlook on the domestic economy in 2012 Q4[1/]

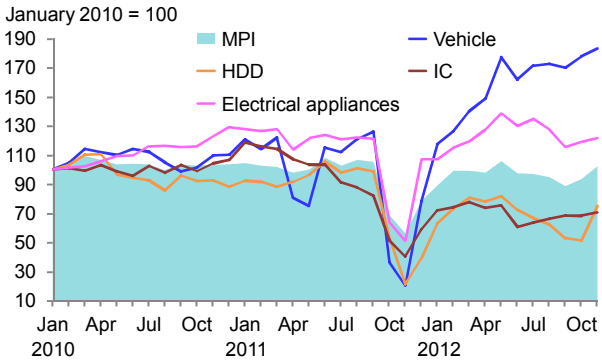
The Thai economy in 2012 Q4 was expected to expand from the previous quarter owing chiefly to sustained robust domestic demand, in private and public consumption and investment. This resulted from government stimulus packages, in particular tax rebates for first-time car buyers, favorable employment conditions, as well as satisfactory business sentiment towards the Thai economy. Moreover, although external demand was still affected by the global economic slowdown, signs of recovery were observed. Production of export-oriented industries was thus expected to improve in the fourth quarter.

Manufacturing production in 2012 Q4 was expected to expand from the previous quarter, both for exports and domestic consumption. Export-oriented industries, especially the hard disk drive industry, displayed clearer signs of recovery consistent with increasing foreign demand. Meanwhile, the integrated circuit and parts industry recovered gradually after the flood problem subsided. At the same time, manufacturing production for domestic consumption continued to expand with domestic

---

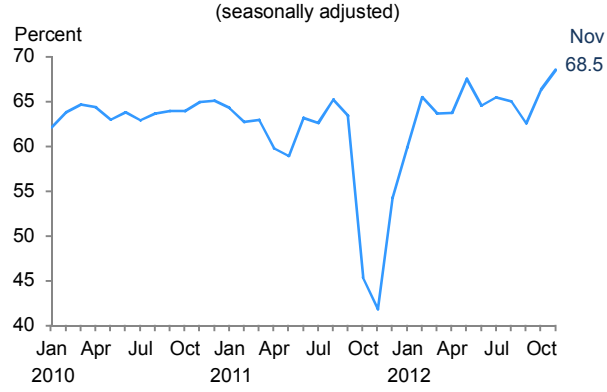
[1/] Economic indicators used in assessing the outlook for 2012 Q4 were computed by the Bank of Thailand, except data on the Manufacturing Production Index and capacity utilization rate, which were computed by the Office of Industrial Economics. The number of tourists and the occupancy rate were, in part, compiled by the Tourism Authority of Thailand. Data on the labor market were obtained from the National Statistical Office. The government expenditure data were provided by the Comptroller General's Department, which compiled by the Fiscal Policy Office, the Ministry of Finance. Economic outlook were obtained from the Economic/Business Information Exchange Program between the Bank of Thailand and business sector.

Chart 2.8 Manufacturing Production Index (MPI)
classified by sectors (seasonally adjusted)



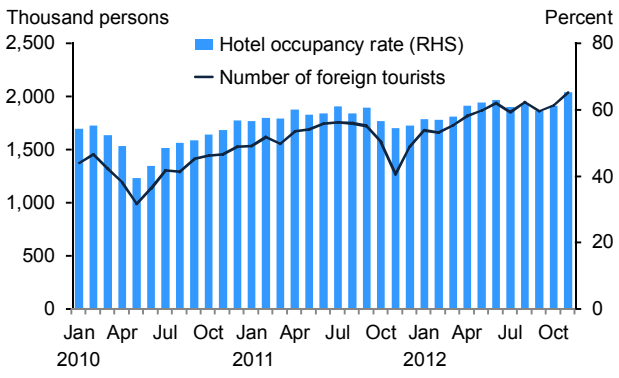
Source: Office of Industrial Economics, Ministry of Industry and seasonally adjusted by Bank of Thailand

Chart 2.9 Capacity utilization rate
(seasonally adjusted)



Source: Office of Industrial Economics, Ministry of Industry and seasonally adjusted by Bank of Thailand

Chart 2.10 Number of foreign tourists and hotel occupancy rate (seasonally adjusted)



Source: Department of Tourism and Bank of Thailand

demand, in particular automobile production which was expected to expand well into the first half of 2013. This was on account of manufacturers speeding up production due to high orders, given incentives from the first-car tax rebate scheme, as reflected by the persistently high Manufacturing Production Index of Vehicles **(Chart 2.8)**. Moreover, overall production of cement and construction material, as well as food and beverages expanded well. Improvements in manufacturing production would likely raise the capacity utilization rate from the previous quarter **(Chart 2.9)**.

In 2012 Q4, the services sector was expected to expand strongly in line with tourism, transportation, trade and construction. The tourism sector grew steadily and was not affected by the global economic slowdown. The seasonally-adjusted number of tourists was expected to increase from the last quarter, especially from the East Asian region. This increase in foreign tourists helped raise the seasonally-adjusted hotel occupancy rate **(Chart 2.10)**. In addition, the transportation and trade sectors were expected to expand well in line with robust domestic consumption. Other services sectors, such as construction, should also expand favorably on the back of increased private investment to support expansion in domestic demand, as well as accelerated disbursement of public investment for mega-projects such as the mass rapid transit system project in Bangkok Metropolitan and its vicinity. The real estate sector, however, was likely to contract from the last quarter due to accelerated investment in the previous periods.

Agricultural production in the fourth quarter was likely to contract slightly from the previous quarter. This was because rice output had increased more than usual during the past two quarters, due partly to price incentives from the government's rice

pledging scheme. Furthermore, cassava production was also expected to decline from the previous quarter due to unfavorable weather conditions in the Northeastern region which destroyed some crops.

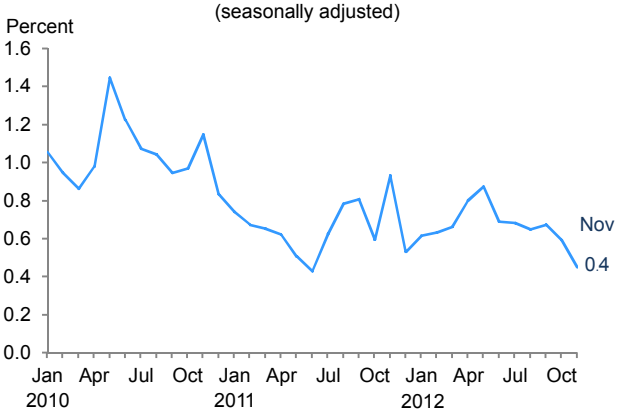
In 2012 Q4, employment was expected to remain stable compared to the preceding quarter, resulting in a persistently low unemployment rate **(Chart 2.11)**. Although agricultural production was likely to contract slightly from the previous quarter, employment in this sector would stabilize at a high level.

For the demand side, in 2012 Q4, private consumption and investment were likely to expand strongly, due partly to government stimulus packages. Although exports were still affected by the global economic slowdown, signs of stability started to be observed.

Private consumption was likely to expand well, as reflected by a rise in the Private Consumption Index in October and November 2012 **(Chart 2.12)**. This was mainly attributable to higher imports of consumer goods, especially food and beverages, coupled with acceleration in spending in the automobile sector which was consistent with the hike in VAT collection.

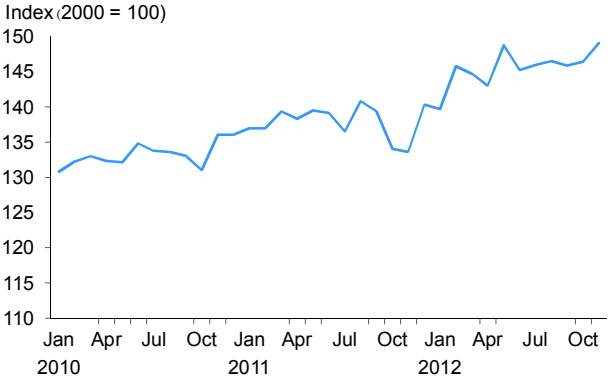
Private investment was expected to grow at a rate close to its normal trend, as observed from a rise in the Private Investment Index in October and November 2012 **(Chart 2.13)**. This was attributable to increased investment in machinery and equipment, as well as in construction to accommodate expansions in domestic demand and flood-damaged repairs still present in some industries. Main factors supporting private investment going forward included strong business confidence in the Thai economy as reflected by rising values of application and issuance of investment promotion certificates by the BOI and high levels of foreign direct investment.

Chart 2.11 Unemployment rate
(seasonally adjusted)



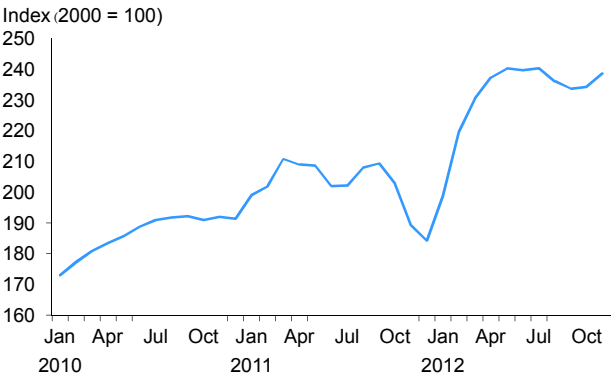
Source: National Statistical Office, and calculations by Bank of Thailand

Chart 2.12 Private Consumption Index (PCI)
(Seasonally adjusted)



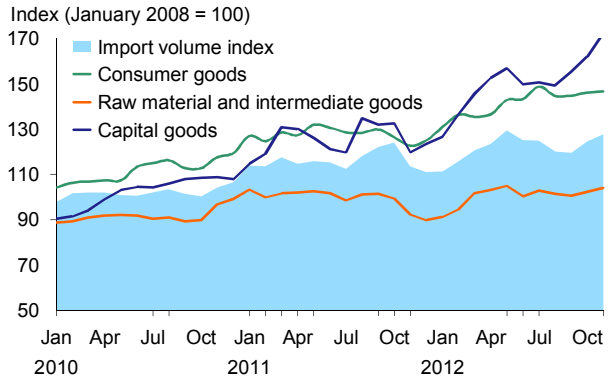
Source: Bank of Thailand

Chart 2.13 Private Investment Index (PII)
(Seasonally adjusted, 3-month moving average)



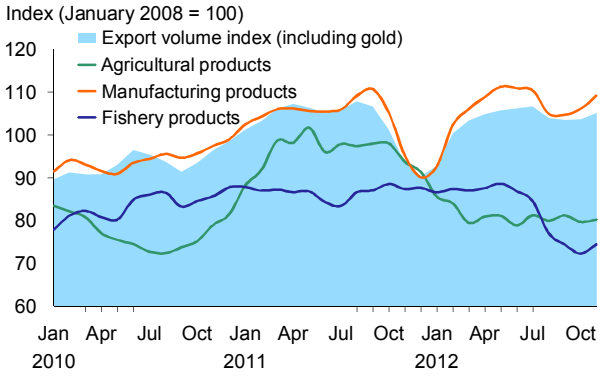
Source: Bank of Thailand

Chart 2.14 Import volume index classified by products
(Seasonally adjusted, 3-month moving average)



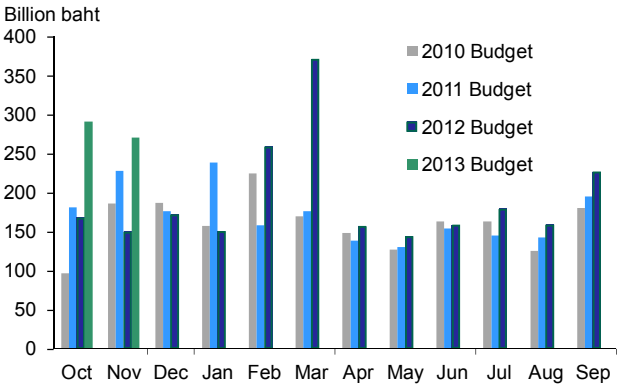
Source: Bank of Thailand

Chart 2.15 Export volume index classified by products
(seasonally adjusted, 3-month moving average)



Source: Bank of Thailand

Chart 2.16 Disbursement of government budget



Note: Excluding principal repayments and replenishments of the treasury reserve
Source: Comptroller General's Department, Ministry of Finance

Continuous growth in private spending and production was likely to lead to expansion in imports of goods and services **(Chart 2.14)**. In particular, imports of automobiles and parts, capital goods, electronic parts, and consumer goods. In addition, imports of services were also likely to expand, particularly transport payments which were expected to increase in line with imported goods.

Although exports were still affected by weak external demand following global economic conditions, there were some signs of stabilization **(Chart 2.15)**. In 2012 Q4, exports of manufactured goods, namely integrated circuits and parts, as well as electrical appliances were expected to slow down. Exports of passenger cars were also expected to decelerate, due to manufacturers' policy to expedite production for domestic customers following high booking. On the other hand, exports of agricultural products were expected to improve slightly from the previous quarter, following a rise in rice exports as exporters hastened to export rice to Nigeria before the tariff hike on rice imports took effect in early 2013. In addition, rubber prices also started to pick up. Meanwhile, exports of services were expected to improve steadily from the previous quarter following increases in foreign tourist arrivals from almost every region, particularly from China and Russia.

The public sector would continue to fuel economic growth **(Chart 2.16)** as reflected in the disbursement of government expenditures in both current expenditures, especially wages and salaries and purchases of goods and services, as well as capital expenditures on investment for the flood restoration and prevention program and investments by the Department of Highways and the Royal Irrigation Department.

Overall, the Thai economy in 2012 Q4 was poised to expand from the previous quarter, largely attributable to growth in consumption and investment, both private and public, as well as in the tourism sector. This was in line with manufacturing production for domestic consumption which was likely to exhibit continuous strong expansion. Meanwhile, merchandise exports and export-oriented manufacturing production started to stabilize.

2.3 Costs and prices

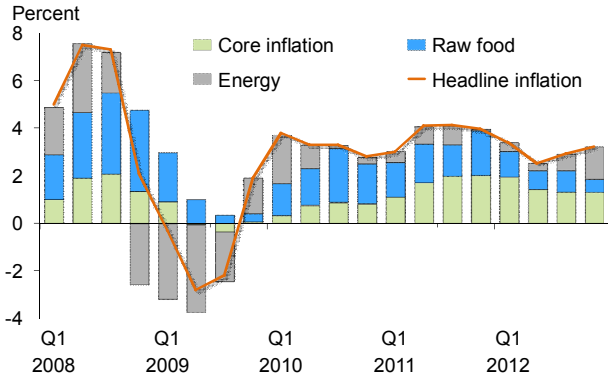
Inflation

Headline inflation in 2012 Q4 stood at 3.23 percent, accelerating from 2.94 percent in the previous quarter due chiefly to a surge in energy prices **(Chart 2.17)**. Energy prices increased by 14.79 percent from the same period last year, accelerating sharply from 7.54 percent observed in the previous quarter. This increase was caused by 1) a slight increase in domestic retail oil prices following global oil prices; 2) a rise in electricity fees for the September to December 2012 period; and 3) a low base effect due to an exemption of oil fund levy last year.

Prices of raw food in 2012 Q4 edged up by 2.94 percent from the same period last year, but slowed down from 4.80 percent in the previous quarter due to a drop in meat and egg prices following an increase in supply. In addition, prices of fruit and vegetables decelerated due to a high base effect caused by last year's flood.

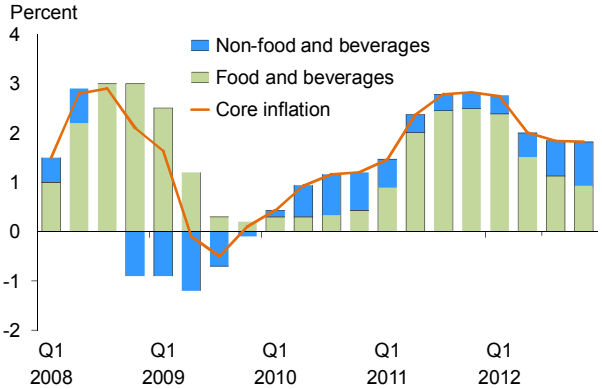
Core inflation in 2012 Q4 stood at 1.82 percent, similar to the previous quarter **(Chart 2.18)**. Prices of food and beverages decelerated in line with prepared food prices which slowed down mainly from

Chart 2.17 Contribution to headline inflation



Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand.

Chart 2.18 Contribution to core inflation



Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand.

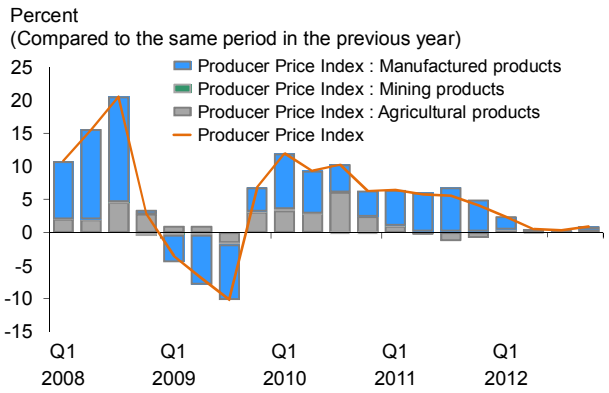
lower costs of raw food. Conversely, prices of non-food and beverages accelerated in line with continuous hikes in tobacco and alcoholic beverage prices due to an increase in excise tax on those items at the end of August 2012. Price acceleration in non-food and beverages also stemmed from a low base effect due to the decline in prices of public transportation services in the third quarter of last year following lower domestic oil prices.

Table 2.2 Quarterly inflation

Unit: Percent	2011	2012				
		Year	Q1	Q2	Q3	Q4
Percentage change from the previous year (%ΔYoY)						
- Headline Consumer Price Index	3.81	3.02	3.39	2.52	2.94	3.23
• Core Consumer Price Index	2.36	2.09	2.74	2.00	1.84	1.82
• Raw food	8.66	4.40	5.78	4.21	4.80	2.94
• Energy	5.59	7.09	3.62	3.02	7.54	14.79
Percentage change from the previous quarter (%ΔQoQ)						
- Headline Consumer Price Index	-	-	0.5	1.2	1.0	0.4
• Core Consumer Price Index	-	-	0.6	0.6	0.4	0.3
• Raw food	-	-	-2.6	3.1	2.2	0.3
• Energy	-	-	7.9	2.4	2.6	1.3

Source: Trade and Economic Index Bureau, Ministry of Commerce

Chart 2.19 Contribution to Producer Price Index



Source: Trade and Economic Index Bureau, Ministry of Commerce, and calculations by Bank of Thailand.

The pass-through from costs to prices in 2012 Q4 slowed down slightly, as reflected by the deceleration of quarter-on-quarter core inflation from 0.4 percent in the previous quarter to 0.3 percent, a level consistent with the 5-year average rate. **(Table 2.2)**

Production costs conditions

The Producer Price Index (PPI) increased by 0.9 percent in 2012 Q4, although accelerating slightly from 0.3 percent in the previous quarter, it still remained low **(Chart 2.19)**, reflecting moderate production costs as a result of the global economic slowdown. In this regard, prices of petroleum and mining products decelerated, while those of

agricultural products accelerated slightly following higher vegetable prices as a result of volatile weather conditions. Prices of manufactured products also accelerated, largely attributable to prices of food, beverages and tobacco.

Inflation pressure in the periods ahead

Overall, inflation pressure was likely to remain stable going forward.

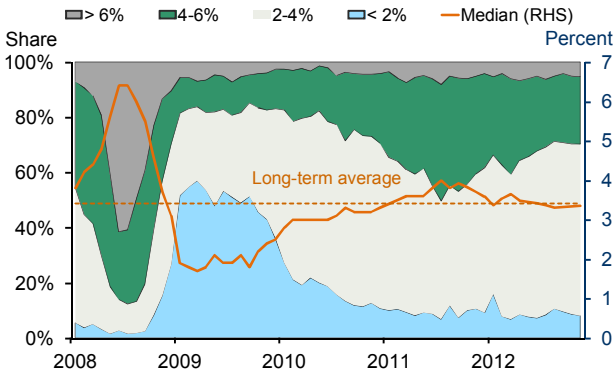
(1) Demand pressure was likely to increase slightly in line with robust domestic demand, particularly in private consumption and investment. Meanwhile, exports were expected to improve steadily in line with gradual global economic recovery.

(2) Inflation expectations remained stable as reflected by the 12-month ahead inflation expectation in November 2012 of 3.3 percent, stabilizing from the previous month and remaining close to its 5-year average **(Chart 2.20).**

(3) Going forward, cost pressures were likely to weaken slightly in line with domestic raw food prices. Although domestic prices of raw food recently increased, it was considered a temporary factor. Meanwhile, world oil and commodity prices were likely to remain moderate.

Risks to inflation that would warrant monitoring going forward include effects of the nationwide minimum wage hike to 300 baht in January 2013, especially psychological effects, as well as the government's plan to restructure energy prices which could be higher than assessed.

Chart 2.20 12-month ahead inflation expectation
(November 2012)

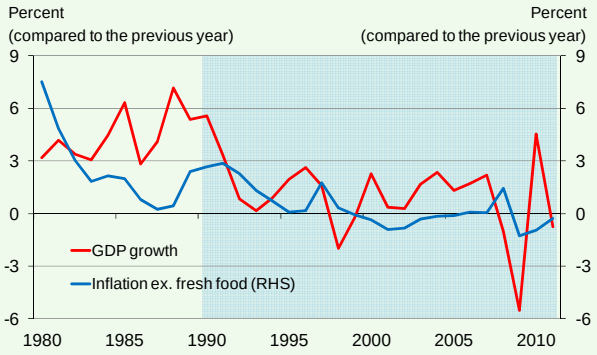


Source: Bank of Thailand's Business Sentiment Survey

Outlook on Japanese economic policy after the December 2012 election

During the past two decades, the Japanese economy has been weak with chronic deflation **(Chart 1)**. Despite various stimulus efforts, for example injection of money into the economic system via government investment and prolonged monetary easing including many years of near-zero policy interest rate and the asset purchase program by the Bank of Japan to inflate the economy, effectiveness has been unclear. Japan has remained weak and has entered into recession many times. Japan is also faced with structural problems, such as aging population, labor shortages and decrease in competitiveness of the manufacturing sector. These factors are key to the economic downturn and have not yet been resolved. During the latter half of 2012, Japan has once again entered a recession following the impact of the euro area debt problem on the global economy, and historical strengthening of the yen.

Chart 1 Japan's GDP growth and inflation (excluding fresh food)



Sources: IMF WEO (October 2012) and CEIC

In the December 2012 general election, the former opposition party (Liberal Democratic Party – LDP) led by Shinzo Abe won a landslide victory. The LDP has then formed a coalition government with the New Komeito party with a two-third majority in the senate, which would ease subsequent budget formulation and policy implementation. In this regard, in the election campaign, the LDP announced various policies to take Japan out of deflation. Notable ones include:

1. Fiscal policy

The new government would stimulate the economy by increasing public spending. It is expected that additional budget of around 7-13 trillion yen, or equivalent to about 1.5-2.5 percent of GDP would be announced in 2013 Q1. A ten-year plan for infrastructure investment is also expected. Furthermore, the government would likely postpone a hike in consumption tax initially planned for April 2014 should economic outlook remain weak, and reduce the corporate income tax rate from the current 38 percent[1/].

2. Monetary policy

The new government would tend to favor greater monetary easing to resolve the deflation problem whereby the medium- and long-term inflation target could be increased from 1 to 2 percent in order to raise long-term inflation expectation. In this regard, Bank of Japan would consider this proposal in its monetary policy committee meeting in January 2013. To achieve the proposed inflation target, analysts and investors expect that Bank of Japan would implement the following additional measures. (1) Conduct an open-ended asset purchase program. (2) Increase

[1/] Japan has one of the highest corporate income tax rates in the world

the tenures of government bonds to be purchased from the present range of 1-3 years. (3) Expand the types of private assets the Bank of Japan can purchase. (4) Establish a public-private joint fund to purchase foreign bonds in order to weaken the yen.

In this regard, should the government succeed in implementing these policies, the short-term outlook of the Japanese economy would likely brighten. However, in the medium- and long-term, the government still faces various threats that could restrict policy implementation as follows.

- The prolonged high fiscal deficit would further increase public debt from the current level of 235 percent of GDP.[2/] Looking ahead, the absence of a plan to reduce public debt would heighten the risk of the economy being down-graded.
- Demand for as well as ability to hold government bonds of the private sector is limited as

  (i) commercial banks' balance sheets would be at risk following a potential loss from holding such bonds should yields increase. At present, Japanese commercial banks' holding of government bond is around 24 percent of total assets, which is high compared to other advanced economies[3/]

  (ii) households' demand for such bonds would likely decline in line with the trend of aging population where incentive to save lessens.
- Risk from reliance on foreign investors for liquidity remains as foreign holdings of Japanese assets normally fluctuate with the economic outlook and investors' confidence. In addition, a deterioration of Japan's fiscal sustainability could also prompt these investors to decrease their holdings of Japanese assets.
- Bank of Japan would likely face greater political pressure from the new government including the change in inflation target. In this regard, most investors view that the target of 2 percent would be too high for Japan (the average inflation rate over the past ten years is -0.2 percent), posing a significant risk of a below-target performance in the medium- and long-term thereby affecting Bank of Japan's policy implementation.

Furthermore, focusing on demand-led stimulus could only boost the economy in the short-term while restrictions remain in the long-term. Therefore, a sustainable solution for Japan would be to address weaknesses in the economic structure, for example, by restructuring the tax and pension systems, resolving the problem of labor shortages, and implementing measures to strengthen the manufacturing sector.

Nevertheless, the government's display of commitment towards fiscal and monetary stimulus measures has caused the yen to weaken. In a way, such development would be beneficial for the export sector and increase Japan's competitiveness. However, many policies still remain unclear both in terms of detail and likelihood of implementation. These factors would determine the outlook of the yen as well as Japan's economic recovery in 2013 and subsequent periods.

[2/] 2011 public debt data from IMF World Economic Outlook (October 2012)

[3/] Commercial banks in the US, the UK, France and Germany hold about 5 percent of assets on average in their respective government bonds (data as of end 2011).

Monetary and Financial Stability

3. Monetary and Financial Stability

Thailand's overall financial stability improved thanks to better sentiment in the financial markets which was partly due to additional monetary easing by advanced economies. Improvement in investor confidence also led to rising investment in the bond and equity markets. Stability in other sectors, namely financial institutions, household, and corporate was also well maintained. While there was no immediate risk to fiscal stability, looking ahead, the ratio of public debt to GDP could rise following the increase in government spending.

Going forward, key risks to be monitored included (1) the impact of monetary easing in the US together with monetary and fiscal stimulus measures in Japan on volatility in Thai financial markets and the Thai baht; (2) uncertainties in the global economic recovery that could affect future corporate and household income; and (3) the persistent high growth in consumer credits, especially auto leasings and other personal loans.

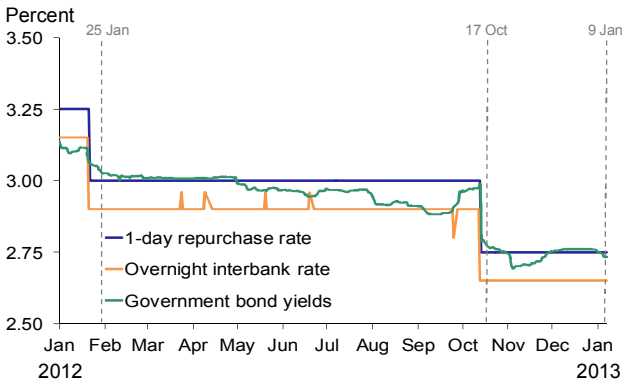
3.1 Financial markets

Interest rates in the money and bond markets adjusted downward in line with the policy rate cut as well as global financial market and bond supply conditions. Investor confidence improved partly as a result of monetary easing measures in advanced economies. Thus, foreign investment in the bond and equity markets increased and led to some strengthening of the baht.

Money and bond markets

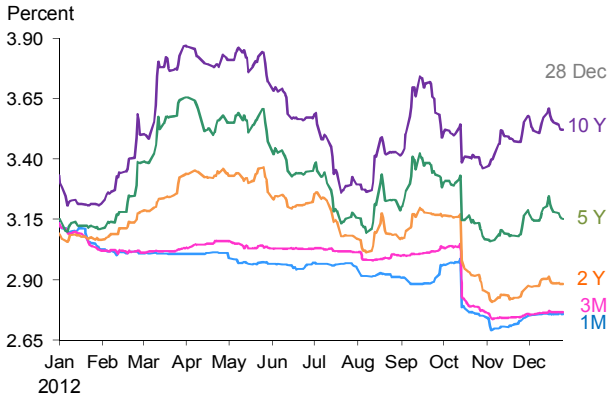
In 2012 Q4, short-term interest rates initially fell then began to stabilize in line with a policy rate cut by 25 basis points to 2.75 percent per annum on 17 October 2012 and subsequent decisions to hold the policy rate on 28 November 2012 and 9 January 2013, respectively **(Chart 3.1)**.

Chart 3.1 Money market short-term interest rates



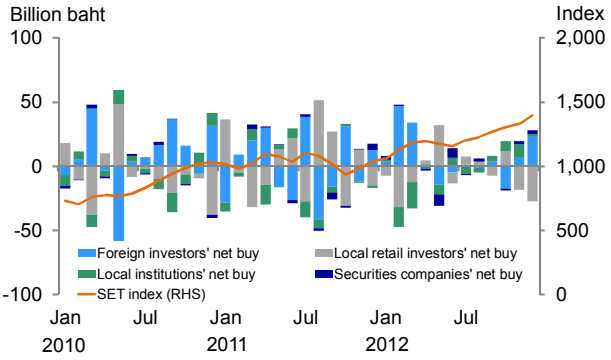
Sources: Bank of Thailand and the Thai Bond Market Association (ThaiBMA)

Chart 3.2 Government bond yields



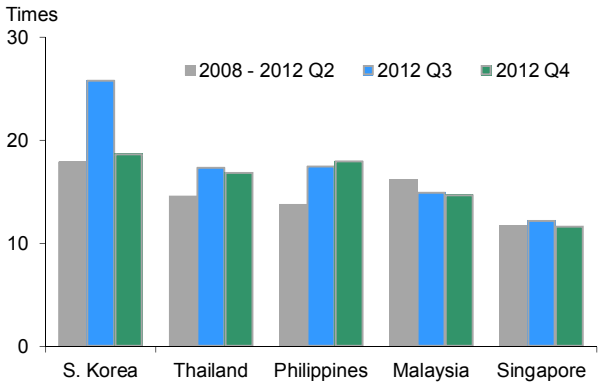
Source: The Thai Bond Market Association (ThaiBMA)

Chart 3.3 The Stock Exchange of Thailand index
(up to 28 December 2012)



Source: The Stock Exchange of Thailand

Chart 3.4 Average Price to Earnings by country



Source: Bloomberg

Overall, government bond yields in 2012 Q4 moved in tandem with global financial market conditions and bond supply. At the beginning of the quarter, bond yields fell due partly to inflows from abroad as well as a policy rate cut on 17 October 2012 **(Chart 3.2)**. Towards the end of the quarter, however, yields on medium- to long-term bonds rose following a larger-than-usual government bond auction in November 2012 together with investors' switch to equity investment in December 2012 as confidence improved after the announcement of further monetary easing in the US.

Equity market

Thailand's stock exchange index in 2012 Q4 continued to increase given investors' confidence in the global financial markets, Thai economic fundamentals and strong corporate profitability. Demand was mainly from foreign investors and local institutions whose net buy was at 3.8 and 19.9 billion baht, respectively. The large increase in institutional investment was due to retail investors' demand for Long-Term Equity Funds (LTF) and Retirement Mutual Funds (RMF) during the end of the year for tax deduction purposes **(Chart 3.3)**.

Looking ahead, the increase in equity prices would be closely monitored. In 2012 Q4, the Price to Earnings (P/E) ratio stood at 16.9, declining slightly from 17.4 in the previous quarter but was still higher than its historical (2008 Q1–2012 Q2) average of 14.6 **(Chart 3.4)**. In addition, uncertainty regarding global economic recovery and the impact of advanced economies' monetary easing measures, such as the US' QE3 and QE3+ and Japan's monetary and fiscal stimulus packages, could cause capital flow and stock market volatility, which would also warrant close monitoring.

Foreign exchange market

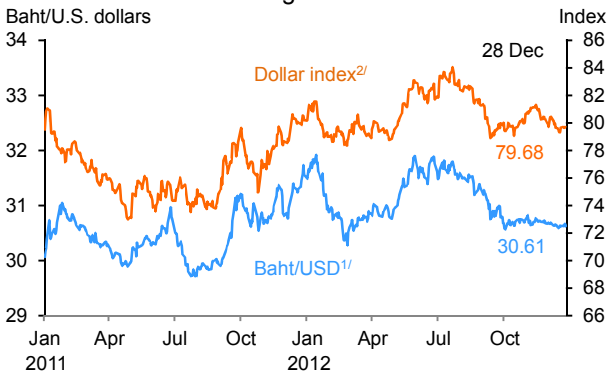
In 2012 Q4, the Thai baht averaged at 30.67 baht per US dollar, strengthening by 2.2 percent from the previous quarter. The movements were quite limited within a narrow range throughout the quarter **(Chart 3.5)**, reflecting greater stability in the global financial markets and more balanced inflows and outflows. Factors that caused the Thai baht to strengthen slightly in this quarter included (1) investors' expectation of a continuation of monetary easing in the US following Obama's election victory (November 6, 2012); (2) likelihood of additional monetary and fiscal stimulus packages in Japan after Shinzo Abe's election victory (December 16, 2012); and (3) better-than-expected Chinese and US economic data.

Nevertheless, the Thai baht weakened against the US dollar at several points during the first half of the quarter as foreign investors slowed investment in Asian equity and bond markets due to uncertainty of the US fiscal cliff, prolonged public debt problems in Europe, and demand for US dollars from gold traders following a continuous decline in gold prices.

Volatility of the Thai baht was at its lowest level in three years **(Chart 3.6)** given its stable movements, reflecting greater stability in the global financial markets and more balanced capital flows.

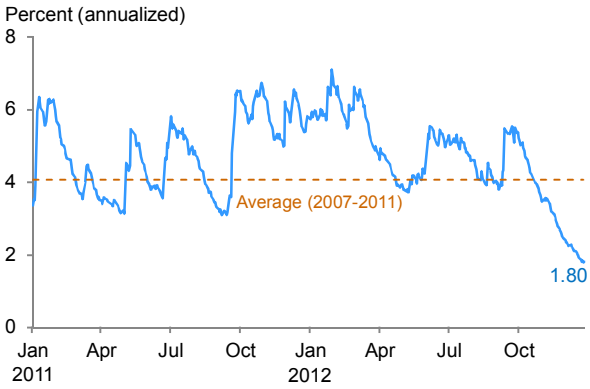
The Nominal Effective Exchange Rate (NEER) averaged at 102.73 in 2012 Q4, strengthening by 1.50 percent from last quarter's average **(Chart 3.7)** due to weakening of major currencies including the US dollar, the Japanese yen and the euro. The Real Effective Exchange Rate (REER) also moved in tandem with the NEER given that relative prices were stable throughout the quarter.

Chart 3.5 Exchange rate and trade-weighted dollar index



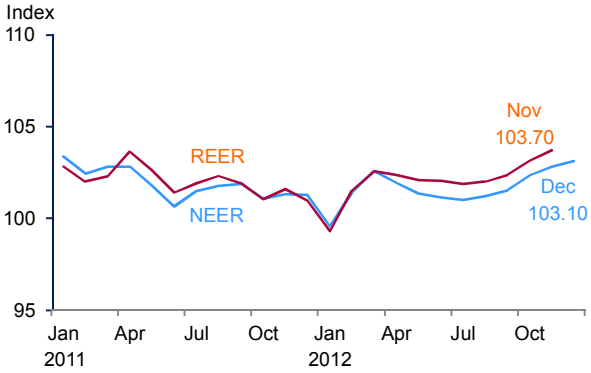
Sources: 1/ Bank of Thailand
2/ Bloomberg

Chart 3.6 Baht actual volatility
(up to 28 December 2012)



Source: Bank of Thailand

Chart 3.7 NEER and REER
(trade-weighted, base year = 2007)



Source: Bank of Thailand

3.2 Financial institutions

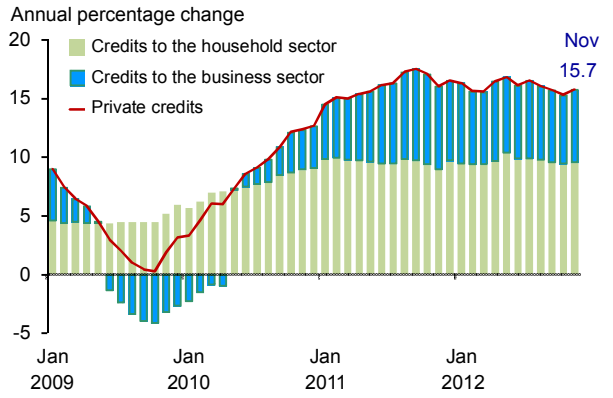
Commercial banks' interest rates in 2012 Q4 declined following a policy rate cut while deposits and credits continued to expand well. Financial institutions stability remained sound overall, though the persistent expansion in consumer loans, particularly other personal loans and the associated loan quality posed risks that would be closely monitored, going forward.

Interest rates, credits and deposits

Commercial banks' interest rates on deposits and loans in 2012 Q4 declined following a policy rate cut on 17 October 2012 after having been stable since the beginning of the year. At the end of 2012 Q4, interest rates on the 3-, 6-, and 12-month deposits of the four largest commercial banks[1/] averaged at 1.86, 2.18, and 2.49 percent per year, respectively. Meanwhile, the MLR stood at 7.00 percent per year. Monetary policy transmission through the interest rate channel in this quarter was fast given a more relaxed liquidity condition, especially of large banks compared to last quarter, thanks mainly to prior competition for deposits through special products.

Private credits at the end of November 2012 continued to expand markedly at 15.7 percent **(Chart 3.8)**. Corporate credit growth remained robust, which could be attributed to the up-cycle in investment. Consumer credits also expanded significantly, particularly other personal loans[2/] and auto leasings that partly stemmed from the tax rebate scheme for

Chart 3.8 Other Depository Corporations' private credits



Source: Bank of Thailand

[1/] Bangkok Bank, Krungthai Bank (KTB), Kasikorn Bank (K-bank) and the Siam Commercial Bank (SCB)

[2/] Other personal loans include personal loans under supervision (non-collateral loans) and personal loans not under supervision.

first-time car buyers. Moreover, the Senior Loan Officer Survey as of end-November 2012 revealed that demand for private credits in 2013 Q1 from both corporates and households would continue to increase **(Chart 3.9)**.

Deposits and bills of exchange (B/E) as of end-November 2012 expanded well at 12.1 percent from the same period last year, partly as a result of the ongoing competition for deposits through special products to replace expiring B/E and support the high growth in private credits.

Stability of financial institutions

Financial institutions' stability in 2012 Q4 was sound, as reflected by the high ratio of capital to risk-weighted assets and strong profitability.

The most recent balance sheets of commercial banks revealed a strong financial position overall with the ratio of capital to risk-weighted assets at 16.0 percent, which was almost twice the 8.5 percent legal requirement. Moreover, liquidity remained abundant, as reflected by continued expansion in deposits and a persistent decline in the loan to deposit and B/E ratio. Commercial banks' net profit of 45.2 billion baht also echoed the sector's strong performance **(Table 3.1)**.

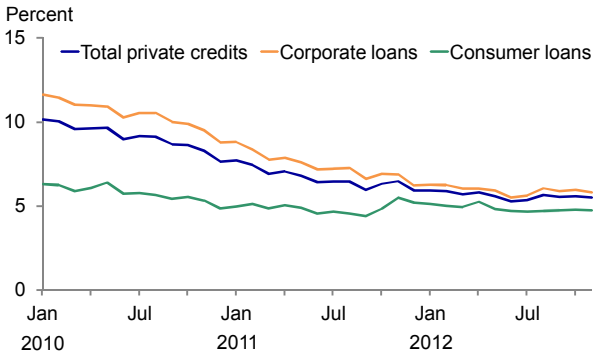
Nevertheless, major risks that could threaten financial institutions' stability in the periods ahead included the persistently high growth in consumer loans, especially other personal loans given a continued rise in their delinquency rate even-though delinquency and NPL ratios remained stable at 5.5 percent. **(Chart 3.10)**

Chart 3.9 Expected demand for corporate and consumer loans in 2013 Q1



Source: Senior Loan Officer Survey

Chart 3.10 Delinquency and NPL ratio



Source: Bank of Thailand

3.3 Non-financial sectors

Overall, stability of non-financial sectors was well maintained as household and corporate outlook improved thanks to recoveries in both domestic and global economies. Nonetheless, risks in the upcoming periods remained from accelerating credits and household debt.

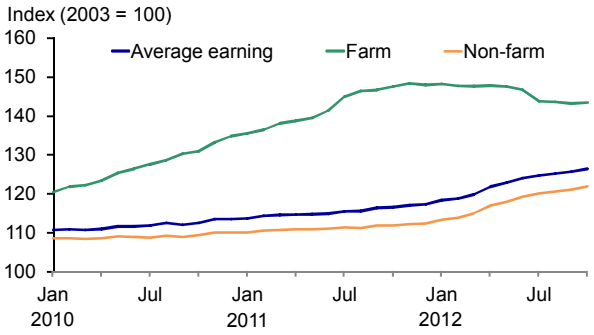
Household sector

Stability of the household sector remained sound overall, as reflected by high employment. However, the expansion in consumer loans, especially auto leasings and other personal loans would be monitored as these could adversely impact households' debt servicing ability, going forward.

Stability of households' income and employment remained robust thanks to high employment, as reflected by a low unemployment rate (seasonally adjusted) in November 2012 of 0.4 percent, as well as strong farm and non-farm average earnings **(Chart 3.11)**. Moreover, a business survey[3/] on the impact of the nationwide 300 baht minimum wage policy on employment conditions revealed that there was no sign of a widespread lay-off as most businesses still experienced labor shortage due to tight labor market conditions.

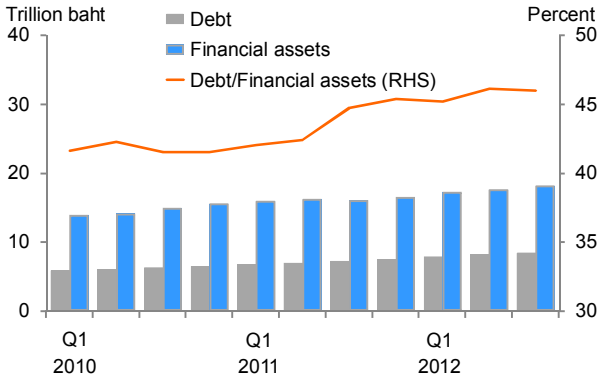
Meanwhile, households' financial conditions remained strong, as evident from liquidity, measured by the ratio of debt to financial assets of households in 2012 Q3, which continued to be low at 45.9 percent **(Chart 3.12)**, and overall loan quality, as reflected by the ratio of delinquent and non-performing loans to total loans, which also remained low at 4.8 percent.

Chart 3.11 Households' income*



Note: *seasonally adjusted, real income (12-month moving average)
Sources: Office of Agricultural Economics, National Statistics Office, calculations by Bank of Thailand

Chart 3.12 Ratio of debt to financial assets



Source: Bank of Thailand

[3/] The Bank of Thailand's Business Liaison Program

Although there was no apparent sign of an imbalance in the household sector, two key risks that would require close monitoring in the following periods included:

(1) the persistent expansion in consumer loans, particularly auto leasings and other personal loans, partly stemming from government stimulus measures, which could adversely impact households' debt servicing ability and loan quality, especially in the low-income group; and

(2) risk of worsening employment outlook, resulting from the impact of global economic uncertainties on the Thai economy as well as the impact of the 300 baht minimum wage policy on employment level.

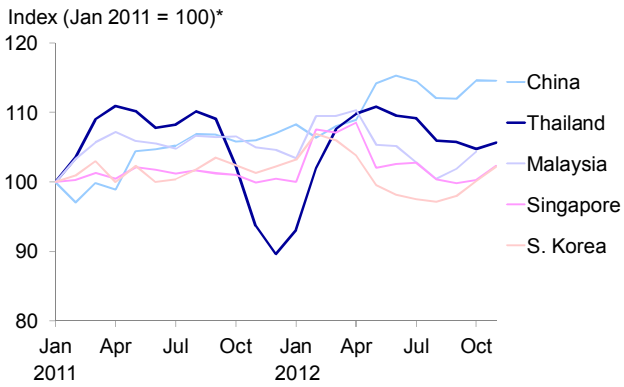
Corporate sector

In 2012 Q3, overall corporate performance remained healthy. However, risks that could affect corporate stability in the periods ahead included the impact of global economic uncertainties on the Thai economy, particularly the export sector, and cost pressures arising from the 300 baht minimum wage policy, which would take full effect in the beginning of 2013.

Overall performance of non-financial companies listed on the stock exchange of Thailand remained healthy. The median of firms' operating profit margin (OPM), which reflected profitability, stood at 7.0 percent, increasing slightly from 6.9 percent in the last quarter. Meanwhile, debt servicing ability and liquidity remained high, as reflected by the median of firms' interest coverage (ICR) ratio of 3.7 and the median of firms' debt to equity (D/E) ratio of 0.9, respectively, which remained stable from the previous quarter.

Looking ahead, corporate profitability was expected to remain robust following expansion in

Chart 3.13 Thailand's export value index



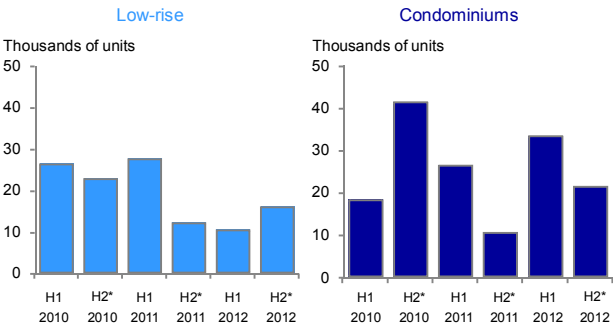
Note: * Seasonally adjusted, 3-month moving average
Sources: CEIC, calculations by Bank of Thailand

domestic demand and a more stable Thai export outlook in line with regional developments **(Chart 3.13)**. However, risks that could affect corporate stability in the periods ahead included the impact of global economic uncertainties on the Thai economy and corporate sector, especially export-oriented industries, and higher cost pressures stemming from the government's wage policy and labor shortages, particularly in the manufacturing and construction sectors.

Real estate sector

The real estate sector experienced some slowdown towards year-end after a continuous expansion from the beginning of the year. Prices in some areas accelerated markedly though no broad-based concern was noted. Factors that needed close monitoring in the next periods included higher production costs and labor shortages which could affect construction projects as well as prices.

Chart 3.14 Number of newly launched residential units in Bangkok and its vicinity



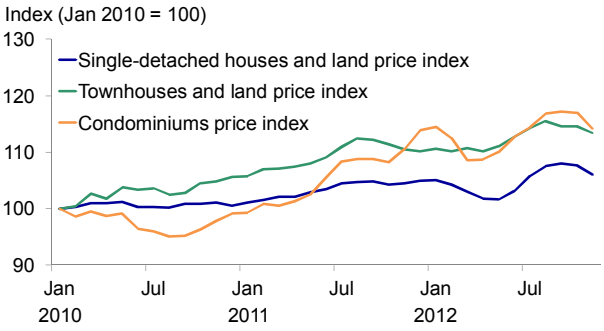
Note: * Total during July to November
Sources: Agency for Real Estate Affairs (AREA), calculations by Bank of Thailand

Thailand's overall real estate activities expanded continuously from the beginning of the year, especially the low-rise housing sector, which improved notably from both the demand and supply sides during the second half of 2012 **(Chart 3.14)**, partly owing to the return of buyers' and firms' confidence following recovery from the flood. Nonetheless, some slowdown was noted towards year-end after experiencing strong acceleration in the previous periods. Housing demand, as reflected by the three-month moving average of the number of approved mortgages from banks (Bangkok and its vicinity) in November 2012 declined from the previous month. On the supply side, firms slowed new openings in Bangkok and its vicinity, especially condominiums, given that a large number of projects had already been launched during the first half of the year **(Table 3.1)**.

Risk of a real estate bubble declined as prices of all types of housings had been stabilizing since the beginning of 2012 Q4 **(Chart 3.15)**. The continuous acceleration in prices from 2012 Q2, particularly for condominiums, was not a threat since it had been in line with rising costs such as higher land prices due to the expansion of public transportation networks as well as wages. Moreover, banks' credit standards were at a normal level and borrowers' debt servicing ability was satisfactory, as reflected by the debt service coverage ratio (DSCR) median of 4.7 in November 2012, which was higher than banks' lending practice of 2.0 **(Chart 3.16)**.

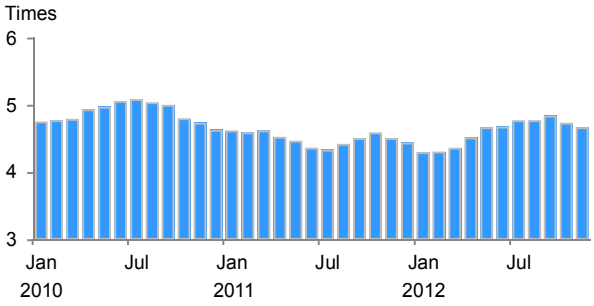
However, risks that required monitoring in the periods ahead included rising costs owing to higher prices of land and construction materials, a wage hike, and labor shortages. All these costs could impact housing construction, prices and foreign investment in the real estate sector in the following periods.

Chart 3.15 Housing price index in Bangkok and its vicinity



Note: New mortgage approvals by commercial banks, calculated by hedonic regression (3-month moving average)
Source: Bank of Thailand

Chart 3.16 Debt service coverage ratio (DSCR)



Note:
(1) DSCR = monthly income/monthly debt payment (principal and interest)
(2) Calculated for new mortgages in Bangkok and its vicinity
(3) 50th percentile (3-month moving average)
Source: Bank of Thailand

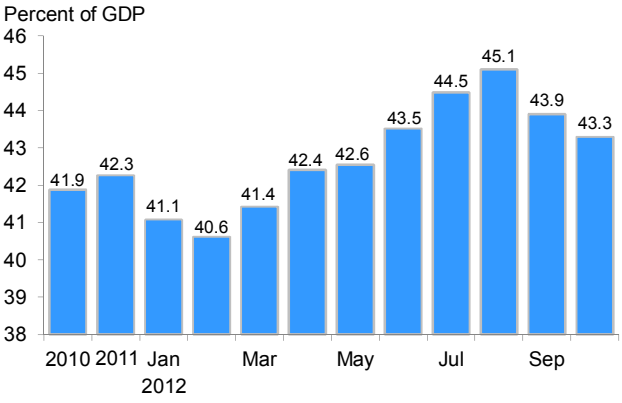
Fiscal sector

Stability of the fiscal sector was well maintained but the ratio of public debt to GDP could increase, going forward, owing to higher public spending.

Stability of the fiscal sector was well maintained, as reflected by the ratio of public debt to GDP as of October 2012 that stabilized from the end of last fiscal year (September 2012) **(Chart 3.17)** and remained well below the threshold for fiscal sustainability of 60.0 percent.

Nevertheless, looking ahead, public spending would likely continue rising from a persistent budget deficit owing to ongoing fiscal stimulus measures, for example the first-car tax rebate scheme, off-balance sheet funding and spending such as the rice pledging scheme, the water management project and

Chart 3.17 Public debt to GDP



Note: Fiscal year
Source: Public Debt Management Office

investment in large infrastructure projects. Meanwhile, revenue collection would likely decline due to tax rate cuts including the decrease in corporate tax rate from 30.0 to 23.0 and 20.0 percent in 2012 and 2013, respectively, and the reduction in personal income tax rate of the highest bracket from 37.0 to 35.0 percent. Such imbalance in government revenue and expenditure could cause public debt to increase, going forward.

Table 3.1 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2009	2010	2011	2012					
				Q1	Q2	Q3	Oct	Nov	Dec
1. Financial markets sector									
Bond market									
Bond spread (10 years–2 years)	2.1	1.3	0.4	0.3	0.5	0.3	0.4	0.6	0.7
Equity market									
SET Index	584.0	849.8	1,025.5	1,120.0	1,168.7	1,228.0	1,298.4	1,292.4	1,391.9
Actual volatility (SET Index)[1/]	26.0	17.6	21.3	13.1	16.7	11.0	9.8	9.0	8.9
Price to earnings ratio (times)	19.9	13.5	12.5	13.9	15.0	17.4	17.1	16.7	17.0
FX market									
Actual volatility (baht)	3.7	3.4	4.8	6.0	4.5	4.7	4.4	3.1	2.1
Nominal effective exchange rate (NEER)	98.5	102.9	101.8	101.2	101.4	101.2	102.3	102.8	103.1
Real effective exchange rate (REER)	102.0	102.9	97.5	101.1	102.1	102.1	103.1	103.7	n.a.
2. Financial institutions sector[2/]									
Minimum lending rate (MLR)[3/]	6.0	5.9	6.9	7.2	7.1	7.1	7.0	7.0	7.0
12-month fixed deposit rate[3/]	0.9	1.0	2.4	2.9	2.9	2.9	2.7	2.5	2.5
Capital adequacy									
Regulatory capital to risk-weighted asset (%)	16.1	16.2	15.1	15.2	15.1	15.9	16.0	n.a.	n.a.
Earnings and profitability									
Net profit (billion baht)	92.2	123.0	147.2	40.1	49.1	45.2			
Return on assets (ROA)	0.9	1.1	1.2	1.2	1.4	1.3			
Liquidity									
Loan to deposit and B/E	85.8	88.3	89.8	90.3	92.4	91.4	91.1	n.a.	n.a.
3. Household sector									
Debt to financial assets	41.3	41.5	45.4	45.2	46.1	45.9			
NPL and delinquency ratio (%)									
Financial institutions :									
Personal loans	6.1	6.0	5.8	5.6	5.8	5.8			
Credit cards	5.4	4.2	4.8	4.6	4.6	4.5			
Thai commercial banks :									
Consumer loans	6.2	4.8	5.2	4.9	4.7	4.8			
Mortgage loans	5.2	4.1	4.0	3.8	3.7	3.8			
Auto leasing	9.6	7.3	8.9	8.1	7.4	7.2			
4. Non-financial corporate sector [4/]									
Operating profit margin (%)	6.6	7.0	3.5	6.8	6.9	7.0			
Debt to equity ratio (times)	0.8	0.8	0.8	0.8	0.9	0.9			
Income coverage ratio (times)	3.9	4.4	2.2	3.6	3.6	3.7			
Current ratio (times)	1.5	1.5	1.5	1.5	1.4	1.5			

Table 3.1 Sectoral Indicators for assessing risks and vulnerabilities to financial stability (cont.)

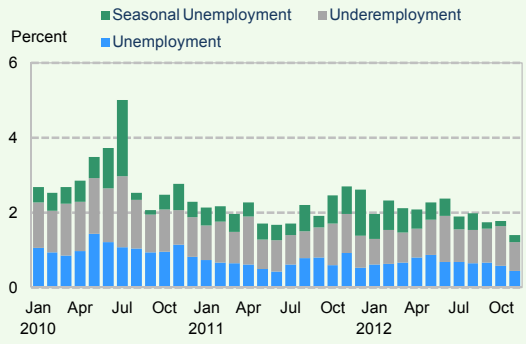
Indicators	2009	2010	2011	2012					
				Q1	Q2	Q3	Oct	Nov	Dec
5. Real estate sector									
3-month moving average of the number of approved mortgages from banks (Bangkok and its vicinity)	49,661	64,134	63,322	12,151	15,723	21,372	6,597	6,013	n.a.
Single-detached and semi-detached houses	15,721	19,176	19,149	3,070	4,208	7,218	2,031	1,654	n.a.
Townhouses and commercial buildings	16,208	21,604	22,102	4,249	5,842	7,550	2,367	2,092	n.a.
Condominiums	17,732	23,354	22,071	4,832	5,673	6,605	2,200	2,267	n.a.
3-month moving average of the number of new openings (Bangkok and its vicinity)	53,749	106,517	92,139	23,236	25,406	25,953	6,834	5,564	n.a.
Single-detached and semi-detached houses	14,081	18,310	21,059	2,303	3,085	3,740	1,354	984	n.a.
Townhouses and commercial buildings	18,319	33,953	25,748	2,407	2,403	7,629	1,396	1,480	n.a.
Condominiums	21,349	54,253	45,332	18,525	19,918	14,585	4,084	3,100	n.a.
Housing price index[5/]									
Single-detached houses (including land)	100.0	100.6	104.9	103.0	103.1	108.0	107.7	106.0	n.a.
Townhouses (including land)	98.9	103.6	110.2	110.7	112.7	114.6	114.5	113.4	n.a.
Condominiums	100.2	99.1	113.9	108.6	112.7	117.2	117.0	114.1	n.a.
Land	100.5	106.9	111.5	117.5	110.5	113.6	115.4	116.6	n.a.
6. Fiscal sector									
Public debt to GDP (%)	43.8	42.5	40.3	41.4	43.3	43.9	43.3	n.a.	n.a.

[1/] Daily volatility (using exponentially weighted average method)
[2/] Based on data of all commercial banks
[3/] Average value of 4 largest Thai commercial banks
[4/] Only listed companies on SET (median)
[5/] Based on data of new approvals by commercial banks using hedonic regression method (January 2010 = 100)

The 300 baht minimum wage policy and its economic impact

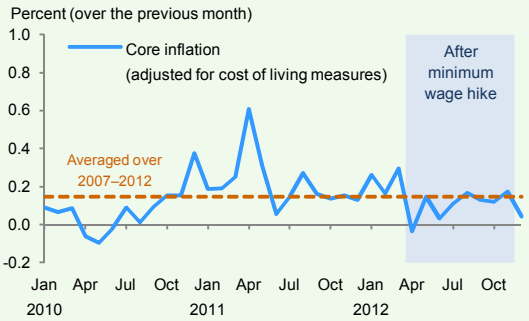
The government started to implement the minimum wage policy hike on April 1, 2012, which led to a 40 percent rise in minimum wage across all provinces. Later on, a second round increase took place on 1 January 2013, raising the nationwide minimum wage level to 300 baht. No further minimum wage increase would be authorized until 2015. Analysis of the first round of increase shows no sign of impact on overall economic stability. Unemployment has stabilized at a low level **(Chart 1)** while the increase in labor cost has neither led to a higher-than-normal pass-through to prices nor created significant inflation pressure **(Chart 2)**.

Chart 1 Unemployment Rate
(seasonally adjusted)



Source: National Statistics Office

Chart 2 Core Inflation
(seasonally adjusted)



Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand

However, the minimum wage hike has led to a shift in the overall earnings structure, causing country-wide private average earnings in 2012 Q2 – Q3 to increase by 17.3 percent from the same period last year – higher than the usual rise of 5.9 percent (taking the average over the last five years[1/]). The increase in average earnings has helped stimulate private consumption and encouraged businesses to improve production efficiency, for example, by substituting labor input with technology. As a result, the country's overall economic potential has risen.

The impact of the first round of minimum wage increase has been trivial due to the following factors:

1. Some businesses have already been paying wage close to the designated level. In addition, the services sector, for example, hotel businesses have been able to include service charge as part of wage.

2. Businesses' overall profit margin after incorporating in the minimum wage increase has remained healthy. Businesses have also been compensated by a corporate income tax rate cut from 30 to 23 percent in 2012. However, taking a micro-level perspective, businesses that have been greatly affected are Small and Medium Enterprises (SMEs), especially those in the manufacturing sector in provinces which had minimum wage less than 250 baht, as well as trading firms (except large businesses in the seven pilot provinces[2/]).

[1/] Using the National Statistics Office's Labor Force Survey during 2004–2012, excluding the global crisis years (2008–2009)

[2/] The seven pilot provinces includes Bangkok, Nakhon Pathom, Nonthaburi, Pathumthani, Phuket, Samut Prakan and Samut Sakhon.

3. Some businesses have been able to upgrade production efficiency and reduce other costs continuously to cushion the impact and maintain profitability. Trading firms in seventy provinces (not including the seven pilot provinces) have formed groups to order products in larger quantities and used different promotion strategies from that of the distribution centers to allow higher turnovers. Meanwhile, manufacturing businesses have adjusted by 1) investing in new machineries to upgrade production efficiency and replace human labor, especially in agricultural processing, plastic and furniture manufacturing firms; 2) improving and streamlining production processes in textile, apparel and electrical appliance manufacturing firms; 3) increasing usage of finished parts, for example, in construction and real estate firms and 4) hiring based on output, for example, in textile, apparel and agricultural processing firms. All these adjustments have helped lessen the impact of the wage hike, thereby allowing businesses to continue operation.

4. The labor market has been tight. Against the backdrop of strong domestic demand, employment has continued to remain high in many industries. Moreover, businesses have been more concerned about labor shortage than higher costs so they have gradually increased wage to the legal minimum level or even higher to attract and retain workers. Despite layoffs in some firms that experienced lower profitability, stronger firms have been able to absorb these workers. Some workers also decided to start their own or return to family businesses. The unemployment rate has thus remained low.

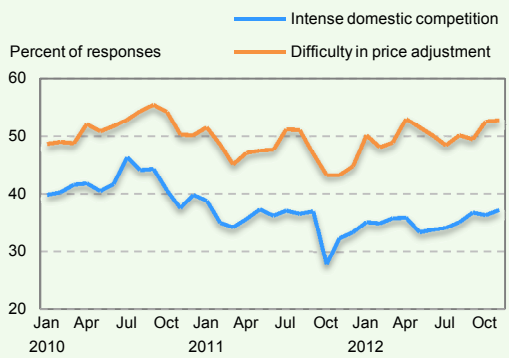
Outlook on the impact of hiking minimum wage to 300 baht in the second round of increase[3/]:

- Outlook on overall profitability. Analysis shows that the second round of minimum wage increase together with a reduction in corporate income tax rate from 23 to 20 percent in 2013 would still affect the same groups, namely SMEs in the manufacturing sector in provinces which had minimum wage less than 250 baht, and the trading sector (except large businesses in the seven pilot provinces). However, under tight labor market conditions and strong domestic demand, labor shortage would still be experienced in other businesses. Therefore, it could be expected that such businesses would be able to absorb these layoffs.

- The impact on overall price stability should also be small as was the case for the first round of increase. Under intense competition, price adjustments would be difficult **(Chart 3)**. Hence, higher labor cost would neither pass-through to prices nor increase inflation pressure significantly.

In any case, the impact of the second round of minimum wage increase would continue to be closely monitored, particularly in the aforementioned risky businesses, SMEs with thinner profit margin and adjustment restrictions, as well as businesses in the seven pilot provinces that might need to increase wages again to retain workers, which could put pressure on wage adjustments and induce further shifts in the overall earnings structure.

Chart 3 Business Restrictions



Source: Business Sentiment Index Survey, November 2012, Bank of Thailand

[3/] The nationwide minimum wage increase to 300 baht would cause businesses in 70 provinces to raise wage between 9.9 to 35.1 percent, averaging at 22 percent. Some provinces such as Phayao, Sisaket, Nan, Tak and Surin would need to hike minimum wage up by 33.7 percent from the previous rate. When combined with the first round of increase (40 percent from the previous rate), these provinces would need to increase wage on average by 86.6 percent in total.